



AQUAscat - ABS System

User Manual

Template AQMANUAL.DOT: 28 December 1995

Document Type	Manual
Document / System Title	AQUAscat - ABS System User Manual
Document Ref:	AQUAscat/0002
Revision:	2.5
Date:	15 October, 2003
Author:	Dave Whitehouse
Checked:	SJC
Approved:	AMS
Classification:	COMMERCIAL IN CONFIDENCE

Revision History

Rev	Date	Details	By	Chk	App
-	19 January 1996	AQ59:C Manual Original Issue	AMS	PVC	AMS
1.00	November 1999	AQ278 Manual Original Issue	AMS	SJC	AMS
1.01	14 December 1999	AQ278 Manual Update Modifications	DW	SJC	AMS
1.02	5 December 2000	User Manual separated from Hardware Manual	DW	AMS	AMS
2.00	19 September 2001	Updated for "AQUAscat"	DW	SJC	AMS
2.1	13 March 2002	Updated for upgrades and new features (Rev 0 ABS Electronics)	DW	SJC	AMS
2.2	2 May 2002	Updated for Rev 1 ABS Electronics	DW	SJC	AMS
2.21	11 December 2002	Update for transducer description packets in file format	DW	SJC	AMS
2.3	23 April 2003	Updated for new receiver gain operation and new features	DW	SJC	AMS
2.4	15 October 2003	Updated for Auxiliary Channels	SJC	DW	AMS
2.5	25 March 2004	Housing specification moved to hardware manual	DW	SJC	AMS
2.5	8 April 2004	Added details for turbidity channel option	DW	SJC	AMS

Acknowledgements

IBM is a registered trademark of International Business Machines, Inc

Windows is a trademark of Microsoft Corporation

Software Warranty

Aquatec Subsea Limited (Aquatec) warrants that the software package described herein will substantially conform to published specifications and to the documentation, provided it is used on the computer hardware and with the operating system for which it was designed. Aquatec also warrants that the magnetic, optical, or electronic media on which the package is distributed and the documentation are free from defects in materials and workmanship for a period of 90 days. Aquatec will replace defective media or documentation, or correct substantial program errors at no charge, provided they are returned to Aquatec within 90 days of the date of delivery.

Except as specified above, Aquatec makes no warranty or representation, either express or implied, with respect to this software package or documentation, including their quality, performance, merchantability, or fitness for a particular purpose.

Because computer programs are inherently complex, and may not be completely free of errors, Aquatec will in no event be liable for direct, indirect, special, incidental, or consequential damages arising out of the use or inability to use the material described herein. In particular, Aquatec is not responsible for any costs including, but not limited to those incurred as a result of lost profits or revenue, loss of use of the computer program, loss of data, the cost of a substitute program, claims by third parties, or for other similar costs. In no cases shall Aquatec's liability exceed the amount of the payment price.

System Warranty

Aquatec Subsea Limited (Aquatec) warrants that the system described herein, termed as the AQUAscat ABS System (AQUAscat), and comprising electronics packages termed the ABS Instrument and ABS Battery, software packages termed AQUAtalk and AQUAview, and other related software products, and various interconnecting cables, will substantially conform to published specifications and to the documentation, with the proviso on the software as described above. Aquatec also warrants that the electronic and mechanical hardware and firmware are free from defects in materials and workmanship for a period of one year. Aquatec will replace defective hardware or firmware, or correct substantial design errors at no charge, provided they are returned to Aquatec within one year of the date of delivery.

Except as specified above, Aquatec makes no warranty or representation, either express or implied, with respect to this system, including its quality, performance, merchantability, or fitness for a particular purpose. Aquatec will in no event be liable for direct, indirect, special, incidental, or consequential damages arising out of the use or inability to use the material described herein. In particular, Aquatec is not responsible for any costs including, but not limited to those incurred as a result of lost profits or revenue, loss of use of the system, loss of data, the cost of a substitute system, claims by third parties, or for other similar costs. In no cases shall Aquatec's liability exceed the amount of the payment price.

Copyright Notice

Neither the software nor the documentation may be copied, photocopied, reproduced, translated, or reduced to any electronic medium or machine readable form, in whole, or in part, without the prior written consent of Aquatec Subsea Limited, except in the manner described in the documentation.

Copyright © 1996, 1999, 2000, 2001, 2002, 2003, 2004 Aquatec Subsea Limited, High Street, Hartley Wintney, Hampshire, RG27 8NY, UK.

Contents

Acknowledgements	i
Software Warranty	i
System Warranty	i
Copyright Notice	i
Contents	ii
1 Introduction	1
1.1 About This Manual	1
2 The ABS Logger System	2
2.1 Introduction	2
2.2 System Overview	2
3 The AQUAscatter ABS System Acquisition Control	6
3.1 Acquisition cycle	6
3.2 Gain Control	7
3.3 The Logging Process	8
4 Software Installation	11
4.1 Introduction	11
4.2 Hardware Requirements	11
4.3 The AQUAtalk Software Package	11
4.4 The AQUAview Software Package	12
4.5 Installation Procedure	12
4.6 USB Driver Installation	13
5 Operation of AQUAtalk Software	14
5.1 Introduction	14
5.2 AQUAtalk Software Overview	14
5.3 Connecting and Disconnecting a Logger	17
5.4 Examining and Changing Logger Parameters	17
5.5 Logger Parameters (Modify Regime)	18
5.6 Recovering Data From the Logger	24
5.7 Deleting Data on the Logger	25
5.8 AQUAtalk Data and Directories	25
5.9 Real Time Data Display	25
6 Operation of AQUAview Software	28
6.1 Introduction	28
6.2 AQUAview Software Overview	28
6.3 General Controls	28

6.4	Program Controls	29
6.5	Menus.....	29
7	Technical Specification	30
8	AQUAscat File Format (Revision 2.4)	31
8.1	Introduction.....	31
8.2	File Version.....	31
8.3	AQUAscat Information	32
8.4	Logging Session Packet	33
8.5	Exception Packet.....	33
8.6	ABS Regime Packet (version 2).....	34
8.7	ABS Magnitude Data Packet.....	35
8.8	ABS Complex Data Packet	35
8.9	ABS Information packets	36
8.10	ABS Transducer Information	36
8.11	Instrument Data.....	36
9	Sample Tutorial	38
9.1	Introduction.....	38
9.2	Session Objectives	38
9.3	Getting Started	38
9.4	Checking and Making Space on the Logger.....	38
9.5	Setting up the Regime	39
9.6	Logging	39
9.7	Recovering Logged Data.....	40
9.8	Viewing Logged Data	40
10	Fault Finding / Correction	41
10.1	Introduction.....	41
10.2	AQUAtalk program will not communicate with logger	41
10.3	AQUAscat did not log when expected	41
10.4	AQUAtalk program will not upload data	41
10.5	Files do not contain all expected data	42
10.6	Logger time is wrong after I have set it and disconnected	42
11	Appendix – Optional External Channels.....	43
11.1	Turbidity and Temperature.....	43

1 Introduction

The AQUAscat ABS System (AQUAscat) comprises acquisition electronics, a data logger, mechanical hardware, cables, software and documentation.

AQUAscat is a stand-alone instrument, which may be connected to the user's computer via an RS232 serial interface (using the RS422 converter supplied) or a USB (Universal Serial Bus) interface. Part of the system is a Windows-based software package (AQUAtalk) that allows the user to communicate with the ABS logger their computer, and a further software package (AQUAview) that allows the user to examine data files produced by the logger.

1.1 About This Manual

This User Manual provides a detailed guide to the ABS hardware and the control software package. Topics covered include:

- ❑ AQUAscat system description
- ❑ The AQUAscat acquisition cycle
- ❑ Installation of software
- ❑ Operation of the AQUAscat control software (AQUAtalk)
- ❑ Operation of the AQUAscat off-line data viewing software (AQUAview)
- ❑ A fault finding guide
- ❑ A description of the AQUAscat data file format
- ❑ A tutorial for a sample deployment

A separate manual – The ABS Hardware Manual – describes implementation specific details of the ABS hardware, including connector details and electronic interface issues.

2 The ABS Logger System

2.1 Introduction

This section describes the main components of AQUAscat in broad, functional terms.

2.2 System Overview

A simplified block diagram of the AQUAscat ABS System is shown in Figure 1. Each of the major system components will be described below.

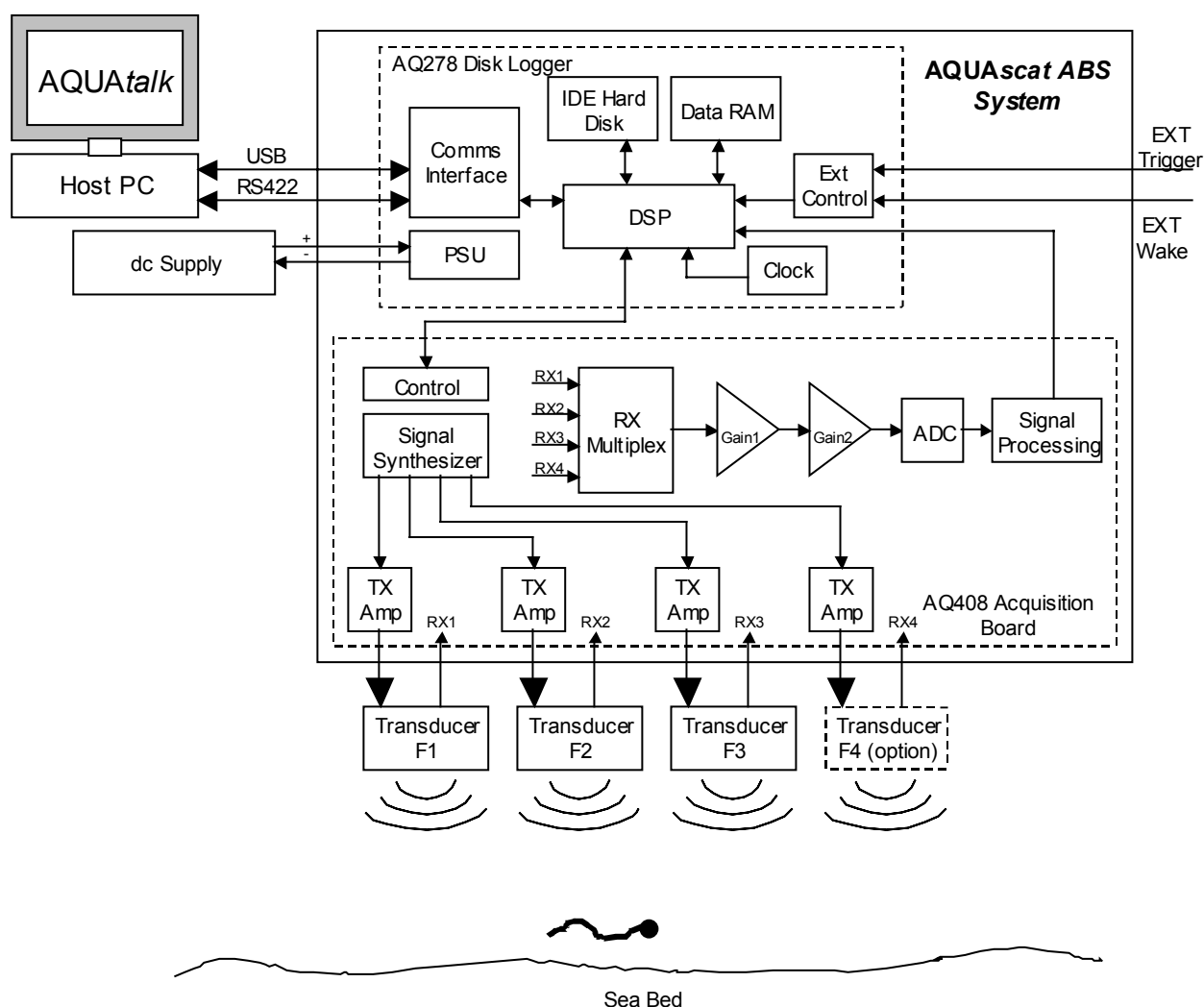


Figure 1 AQUAscat ABS System Diagram

2.2.1 The AQ278 Logger

The logger consists of an electronic printed circuit board and a hard disk drive. The board is based on an Analog Devices ADSP2185 Digital Signal Processor (DSP). This carries out control, signal processing, communications and storage functions.

2.2.1.1 Data RAM

The ABS process generates large amounts of data very rapidly. This data is temporarily stored on the RAM on the logger board, before being permanently stored on the hard disk. The rate at which data fills the onboard logger RAM is dependent on the profile rate and the amount of averaging selected – high rates and minimal averaging generate vast quantities of data very quickly. Once the onboard memory reaches a set threshold it begins writing data to the disk, while new data is still being received – this removes the restriction of the 2Mbytes of logger memory allowing large files to be acquired (i.e. long burst lengths can be acquired at high profile rates, if required).

2.2.1.2 Hard Disk

The hard disk provides the non-volatile storage required to hold the enormous quantities of data that the ABS can generate over the deployment period (which can be several months).

2.2.1.3 Communications Interface

The two interfaces to communicate between the AQUAscat ABS System and the host PC are RS422 and USB. Both interfaces can be used to configure and upload data, but they have different benefits depending on the application.

- ❑ The USB interface is for short range communications where the PC can be located within two metres of the system. It has the significant advantage of operating at much higher data rates than the RS422 interface, which is especially important when uploading data following a deployment. USB support requires Windows 98 / ME / 2000 or higher.
- ❑ The RS422 interface can operate at extended ranges (hundreds of metres between the PC and the system) although at significantly slower data rates (up to 115k baud).

2.2.1.4 Power Supplies

All of the power supplies required internally are derived from the single dc external supply (refer to Hardware Manual for specification). Its circuitry includes controls to switch the power rails on or off, according to external stimuli or a real time clock.

2.2.1.5 External Control

Two signals with a common ground may be used to provide an external logging trigger for the system. These inputs are optically isolated from the main logger electronics, and are designed to operate from 5V logic signals. One signal is connected to the power-up circuitry of the logger (EXT Wake). The logger firmware also interprets this as a trigger to commence logging. The other external signal (EXT Trigger) is for synchronising the AQUAscat to an external trigger.

2.2.1.6 Timing

The logger includes a real time clock to time stamp data. An additional low power timing controller is used to generate regular timing intervals for unattended logging. Precision interval timers are used to set up an accurate sampling regime.

2.2.2 The AQ408 Acquisition Board

The AQ408 Acquisition board contains all of the functions required to generate the precise timing pulses and high frequency transducer drive signals. It also contains several stages of filtering and

amplification along with signal digitisation and processing before being fed back to the logger for processing and storage.

2.2.2.1 Signal Synthesiser

The standard AQUAscatter is supplied with 3 transducers which are each tuned to a set frequency, nominally 1, 2.5 and 5MHz. A fourth transducer can be fitted. These are driven with a pulse of sinusoid for a period proportional to the required distance resolution (typically 1cm). However the drive circuitry is capable of generating almost any signal, as required by the application. Contact Aquatec for further information.

2.2.2.2 TX Amp

Each AQUAscatter is capable of driving up to 4 transducers. Each channel has its own power amplifier to drive nominally 1 watt of power into the transducer for the duration of the transmit pulse.

2.2.2.3 Transducers

The transducers are supplied on two meters of cable unless requested otherwise. Each transducer is tuned to a certain frequency and before deployment the user should ensure that the correct frequency transducer is attached to the correct connector on the housing. In normal operation the transducers are fired for a short period at their tuned frequency and the return echo is monitored on the same transducer. Each of the channels is operated in turn.

2.2.2.4 Gain 1

The user can configure the receiver gain in the AQUAtalk application, to give the best signal for the observation environment. This gain is physically distributed between two gain stages. The Gain 1 stage is a variable gain stage that is configurable by the user from the AQUAtalk application. The stage has a fixed component and a time-varying component to compensate for the increased attenuation of the return echo as the backscatter source becomes further from the transducer head. Abnormally large return echoes may saturate this stage, so the user should carefully select a gain profile that is compatible with the observation environment. A threshold detector is present after this stage, so if the signal does exceed the maximum level, on the real-time display screens a message appears to inform the user of this. If this does occur, there are two options: move the transducer head further from the target, or reduce the gain in AQUAtalk.

2.2.2.5 Gain 2

The Gain 2 stage is constant for the duration of the return. If the transducer is placed in a confined space or very close to a good acoustic reflector (e.g. the seabed) the received signal from the transducer can be large enough that the output from this stage will saturate the following stage, which is the input to the analogue to digital converter. This event will appear on the real-time display screen. In the event of this occurring, the transducer should be positioned further away from the reflecting surface, or the gain should be reduced. If a large gradient is specified for the gain profile in the AQUAtalk saturation may occur towards the end of the profile – in this event, reduce the gradient. In totally clear conditions (i.e. no suspended sediment) saturation can occur even at relatively big separations (up to approximately a meter) if the transducers are directly targeted at a solid surface (e.g. the bottom of a test tank).

2.2.2.6 ADC and Signal Processing

To acquire the high frequency signals received from the transducers, an analogue to digital converter (ADC) is run at a very high speed to over-sample even the highest frequency transducer signal. This high data rate signal is processed to give a lower data rate signal proportional to the concentration within each bin (typically 1cm). This data is fed back to the DSP, which performs the averaging on the data before storing it.

2.2.2.7 External Channels

Slow rate additional channels can be added to the AQUAscat system as options (for example temperature, turbidity and external analogue inputs). The electronics for these channels are located on the PCB in the AQUAscat that holds the hard-disk.

3 The AQUAscat ABS System Acquisition Control

The AQUAscat ABS system has numerous parameters that must be configured prior to successful deployment. The AQUAtalk application automatically configures many of these options, and many of the user interface input options are verified prior to being downloaded to the system. This section describes most of the important features of the system. The actual configuration controls for each parameter are described with the rest of the AQUAtalk application features in Section 5.

3.1 Acquisition cycle

Each acquisition cycle of the AQUAscat ABS System generates a profile. One profile is generated for each enabled frequency. A typical acquisition cycle for a 1 cm resolution, 3 frequencies enabled cycle is shown below:

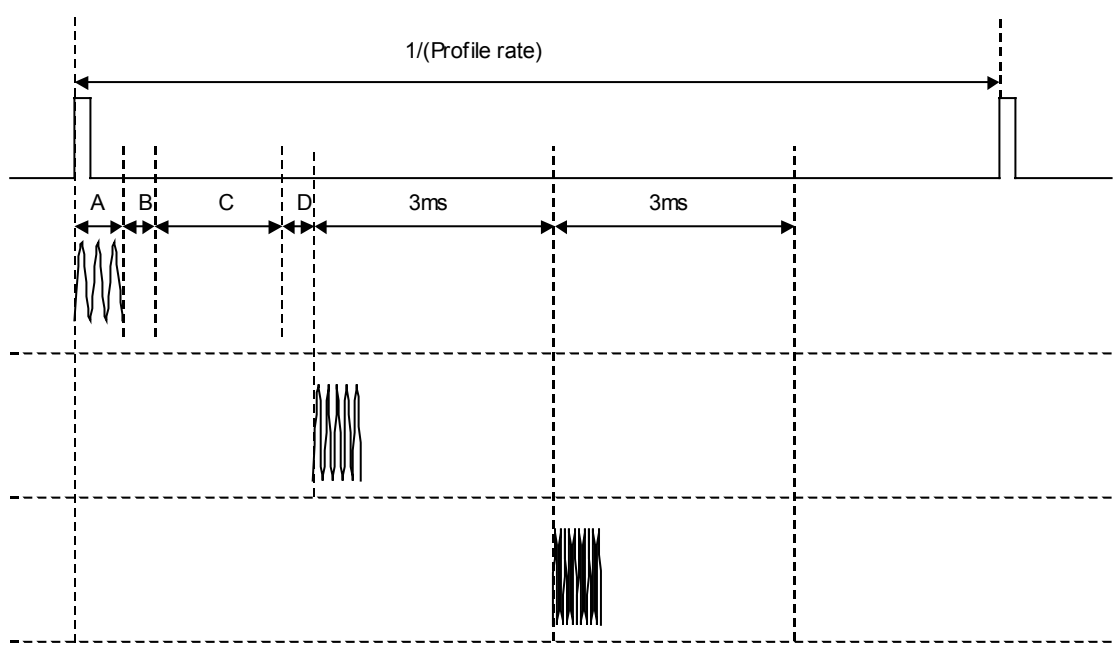


Figure 2 – AQUAscat Acquisition Cycle

The parameters above are described below:

- ❑ A – Transmit pulse length. This period is how long the transmitter drives the transducer for. The pulse length is defined by the required distance resolution.
- ❑ B – Start of receive window delay. Once the transducer has fired, a small delay is required to allow the transducers to recover after the transmit pulse. To allow for this in AQUAtalk the profile start distance is checked by the program to ensure sufficient time is allowed for the effect of the transmit pulse to decay. This distance can be extended, to reduce the amount of data to be stored on the logger. For example if the main object of interest is the sediment very close to the seabed and the transducer is positioned relatively high above it, the start of the receive window can be moved further away from the transducer head to reduce the amount of data acquired.
- ❑ C – Receive window. This is the region that is actually stored on the logger.

- D – Processing. After the receive window has completed there will be a small delay before the next frequency transmit pulse while the final data from the receive period is processed.

The cycle is repeated for each frequency enabled. The total time for each frequency is set at 3ms. After all of the enabled transducers' acquisition periods have finished, the system enters a low power state before the next profile clock. So for a 25Hz profile rate with 3 frequencies active the idle time will be $(1/25\text{Hz} - 3 \times 3\text{ms}) = 31\text{ms}$.

3.2 Gain Control

During each receive pulse at each frequency, the gain on the receiver can be varied with time to account for the increased attenuation of the transmit pulse by the water. The profile is defined in the AQUAtalk application, and can be changed by the user according to the observation conditions.

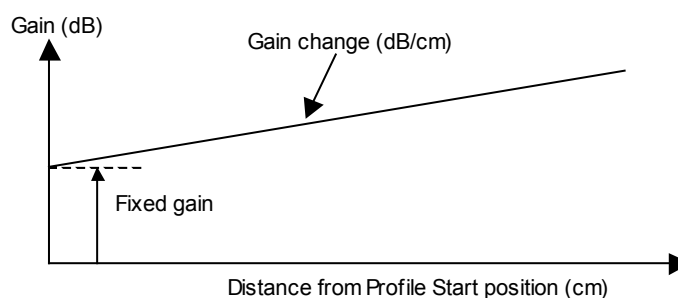


Figure 3 Gain Profiles

If the gain change is set to 0dB cm^{-1} it is held constant at the Fixed Gain value (set in AQUAtalk) for the duration of the receive period. It is then the responsibility of the user to correct for the attenuation in the water by post-processing the data.

If a linear ramp is selected (linear in **dB**) the gain is increased at the set rate (in dB cm^{-1}). The gain will actually change once per bin, so for example if a gain change of 0.01dBcm^{-1} and the resolution is set to 5mm, the gain will change 0.005dB per bin; if the gain is set the same and the resolution is 8cm bins, the gain will change 0.08dB per bin. The following should be noted:

- The start of the gain profile is coincident with the start of the receive window
- If a very shallow gradient is selected the gain may change less frequently than once per bin
- The gain stage in the logger is physically a cascade of two gain stages – one fixed for the duration of the receive window and the other varying for each bin. Therefore if a high initial gain is selected and a steep gain gradient, the sum of the two may exceed the total gain possible with the two stages – the AQUAtalk application will indicate that the maximum gain has been reached and at what distance from the transducer this will occur.

An important issue that the user should be aware of is saturation in the receiver. As the receiver consists of several cascaded stages, if the output of any stage saturates the input to the following stage the data acquired by the AQUAscatter ABS system will be invalid. A detailed view of the gain stages is shown in Figure 4:

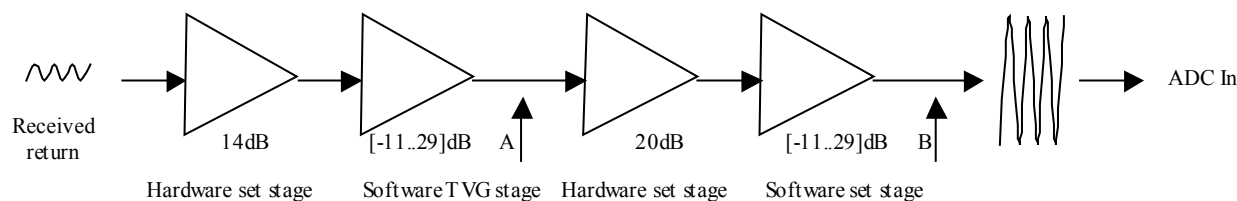


Figure 4 Receiver Gain Stages

The first stage is a very low noise preamplifier. It is link selectable to 14dB or 20dB – the standard system ships with this link (LK11) fitted to provide 14dB of gain. The next stage is the time-varying gain stage, which can change during the receive period. A threshold detector is located at point A to detect saturation. At point A in Figure 4 the peak voltage should be less than $\pm 200\text{mV}$ to avoid saturating the input of the next stage. Point A corresponds to JP12 or link LK3 on the AQ408 acquisition board. The next stage is a fixed gain stage of 20dB. Following this is the final stage which drives into the ADC. At point B a second threshold detector is present to detect saturation of the ADC input. The AQUAtalk application implements a gain distribution algorithm to distribute the gain between the two software configurable stages to minimize system noise.

3.2.1 Recommended Gain Settings

These recommendations are intended as a guide only. The AQUAscat system was set up in a tank of clear water, with the transducers positioned perpendicular to the bottom of the tank. The bottom of the tank consisted of 5mm of level sand. The transducers were at a distance of 1.0m from the bottom of the tank.

To avoid saturation the gain settings were: -

- ❑ 1Mhz Transducer 20dB
- ❑ 2.5Mhz Transducer 35dB
- ❑ 5Mhz Transducer 45dB

3.3 The Logging Process

The following paragraphs describe the detailed activities of logging. The range of parameters, and the possible values to which they may be set, give a general indication of what the logger may be programmed to do.

3.3.1 Preparing to log

Using the AQUAtalk application the regime to be used by the AQUAscat ABS system is configured. Once all of the required settings have been made, the user is prompted to “Write Setup to logger?” – if OK is selected the new settings will be written over the old setup. Once the regime has been downloaded, the user should disconnect from the AQUAscat ABS system – within one minute the system will power down and be waiting for its first trigger (either external logic trigger, or internal timer trigger). The user should note that logging is NOT ENABLED UNTIL THE LOGGER HAS BEEN DISCONNECTED and has POWERED DOWN. If the AQUAscat ABS system is running the USB port of the host PC, it must be unplugged as well as disconnected via the AQUAtalk application, to allow the unit to power down.

3.3.2 Triggering the Logging Process

The AQUAscat ABS system is generally in a state where it is prepared to receive and act upon a timing trigger, which will cause it to begin sampling data. The AQUAscat ABS system is able to accept two different forms of logging trigger, dependent upon the setting of an internal parameter.

If the Auto-logging parameter is enabled, the logger will be triggered from its internal timer. Depending on the setting in the AQUAtalk application the logger will either start on the first alarm from the periodic internal timer, if Instant Start has been selected, or from the logger real-time clock at the time setup by the Start At option. If it is disabled, the logger will not receive regular stimuli and must rely on an external trigger source to initiate logging.

When AQUAscat receives a trigger, it investigates the source to determine what action to take (e.g. a logging trigger or communications connection). If it was previously powered down, it will be powered up on receipt of the trigger, and once initialised will take appropriate action. If the logger was already logging when the trigger arrived, it will not interrupt the logging burst. The exception to this is if the user attempts to connect serial communication. The AQUAtalk application will ask the user if they wish to stop the current logging session, or allow it to complete and connect later.

If the trigger was from the external wake line, the logger uses the *falling* edge of the signal as its timing reference. This may occur up to 50 seconds after the initial wake up. However, if the logger is woken up from a powered down state, at least 8 seconds should be allowed to establish an accurate timing reference.

If the trigger was from the internal timer, the logger starts logging within 8 seconds from the original wake up.

The logger is programmed always to log data when triggered in an auto-logging mode. However, external trigger modes are dependent on a further logger parameter - the 'Log every x triggers'. By setting the 'Log every x triggers' to 3, only every third trigger will result in data being recorded.

3.3.3 Processing and Storing the Sampled Data

All bin samples at each frequency are stored temporarily in memory prior to processing. Depending upon the chosen processing mode, and the data handling parameters, data from a number of consecutive samples is averaged, and may be combined in various groupings, before being stored as required by the frequency and range bin selection criteria. Data is only written to disk at the end of the sampling burst, unless the amount of data in the burst exceeds the loggers 2Mbyte memory capacity in which case the disk will power up and write the current data in memory to disk and power down until the memory is full again or the end of the burst is reached.

3.3.4 Logger Configuration

The current set-up of the logger may be viewed on screen or edited using a wizard to lead you through the tasks. These parameters will be stored by the logger, for future reference when it is logging data.

3.3.5 Logging Sessions

A logging session is the period between commencing logging and terminating logging. Throughout that period, the logging parameters previously set up govern the behaviour of the logger. On starting logging, the logger is set up with the currently selected configuration, and immediately waits for its first logging stimulus.

3.3.6 Data Transfer

When data is to be analysed, it must be recovered from the logger. AQUA*talk* initiates a high speed data transfer across the RS422 serial link (for long distance operation) or USB (for short distances), using a packet-based file transfer protocol to ensure error-free transmission. The USB operates at *significantly* higher data rates than the RS422 is capable of and is strongly recommended for uploading data.

4 Software Installation

4.1 Introduction

This chapter describes the procedure for installing the *AQUAtalk* and *AQUAview* software packages on to a computer. It addresses:

- ❑ Hardware and software environments
- ❑ Description of the package contents
- ❑ Backup routine
- ❑ Installation routine
- ❑ Preliminary check routine
- ❑ System customisation
- ❑ USB Driver Installation

4.2 Hardware Requirements

4.2.1 Computer Compatibility

The *AQUAtalk* and *AQUAview* packages have been written for use with an IBM PC or upwardly compatible machine, running a 32-bit version of Windows (98, NT4, ME, 2000). The software is supplied on a CDROM, but is available in other formats if required.

Note that these are minimum requirements. With the basic configuration described above, the program may be limited in data transfer execution speed. The minimum recommended processor/speed combination is a Pentium 200MHz processor for highest serial data transfer rates. Faster processors may be necessary if USB data upload is to be used.

4.2.2 Serial Communications

There are two methods of serial connection to the system:

- ❑ Via a four wire RS422 connection, with two wires each for transmission and reception. RS422 is not provided as standard on most machines, so a protocol converter is supplied. Alternatively, an add-on board may be fitted to the PC. The software may be configured for COM1 through to COM4. The recommended serial protocol converter is a KK Systems K2 converter, which is supplied with the system.
- ❑ Via a USB interface which is supplied as standard on virtually all current PCs. A cable is supplied to connect directly from the host PC USB port to the logger.

4.3 The *AQUAtalk* Software Package

The *AQUAtalk* package comprises a user manual, incorporated as part of this User Manual, and a CDROM containing the software. *AQUAtalk* is used for all communication with the logger.

4.4 The AQUAview Software Package

The AQUAview package comprises a user manual, also incorporated as part of this User Manual, and is supplied on the same CDROM as the AQUAtalk software. AQUAview is used for examining logger configuration for a given file, and viewing data in a simple graphical format.

4.5 Installation Procedure

4.5.1 CD Installation Procedure

The software is supplied on CDROM. If AutoOpen is enabled on your PC, the browser page CDROM.htm will be loaded. This provides allows you to install both software packages separately, and also to view other Aquatec data sheets:



Press the Install for the package you want to install and follow the on screen instructions.

If your computer does not have CD AutoOpen, you can load CDROM.htm manually from the File-Open menu. If your computer does not have browser support, you can access the installation programs directly in the following locations:

D:\AQUAtalk\setup.exe
D:\AQUAview\setup.exe

Where *D* is your CDROM drive.

4.5.2 Floppy Disk Installation Procedure

If you wish to install on a machine that does not have a CDROM or access to a networked CDROM, then the installation files may be copied to floppy disk. For each application, the first disk should have *setup.exe* and *application.001* on it. The second should have *application.002*, etc. Run *setup.exe* from the first disk and follow the on screen prompts.

4.5.3 File Locations

The software is written using LabWindows CVI. The run-time DLL should be placed in your Windows System directory.

The installation procedure suggests placing the other program files in the folders:

C:\Program Files\Aquatec\AQUATalk
C:\Program Files\Aquatec\AQUAView

although you can select your own preferred location.

A *data* folder is also created in the *Aquatec* folder as the default location for uploaded data files. A sample data file of rather benign acquired data is included to allow *AQUAView* to be demonstrated.

4.6 USB Driver Installation

The USB installation is controlled by the Plug and Play Manager in Windows 98/2000. The logger must be connected to the PC using the supplied USB cable and have a suitable power supply attached.

On connection of the USB cable Windows will automatically detect the ABS logger and try and load a valid driver. If a driver was not previously installed then the "Add New Hardware Wizard" will appear.

When asked, choose to search for the best driver and select the Driver directory on the CDROM drive option in the next dialog box. The driver should then be detected and installed into the system. You will only need to carry out this operation the first time you connect the *AQUAscat* ABS System to the PC. At the end of the procedure you will be prompted to restart the PC.

Once the PC has been restarted the *AQUAscat* will appear within the Control Panel's System dialog under the *AQUAlogger* class type in the Device Manager tab. If this is not present then the logger is either not connected, not powered or a software fault has occurred. A red cross over the *AQUAlogger-AQUAscat System* symbol indicates a problem with the driver.

Once the driver has been installed the user connects to the logger from within *AQUAtalk* in the same way as for an RS422 connection.

5 Operation of AQUAtalk Software

5.1 Introduction

This section covers the general operation of the AQUAtalk software. As this is the user interface for the whole data logging system, it includes detailed descriptions of logger parameters, logging modes, and general operation details. In particular, this section discusses:

- ❑ The user interface
- ❑ The basics of logger connection
- ❑ Logger parameters
- ❑ Logging sessions
- ❑ Data recovery
- ❑ Miscellaneous features

5.2 AQUAtalk Software Overview

AQUAtalk has been written to run under 32-bit Windows (98/2000/ME/NT) on an IBM PC or compatible. The software is designed to provide the user with a self-explanatory method of command entry, using simple keystrokes or mouse movements. The risk of entering incorrect information is reduced by the software's entry verification routines, which warn of and prevent invalid or unsafe data entry.

5.2.1 General Controls

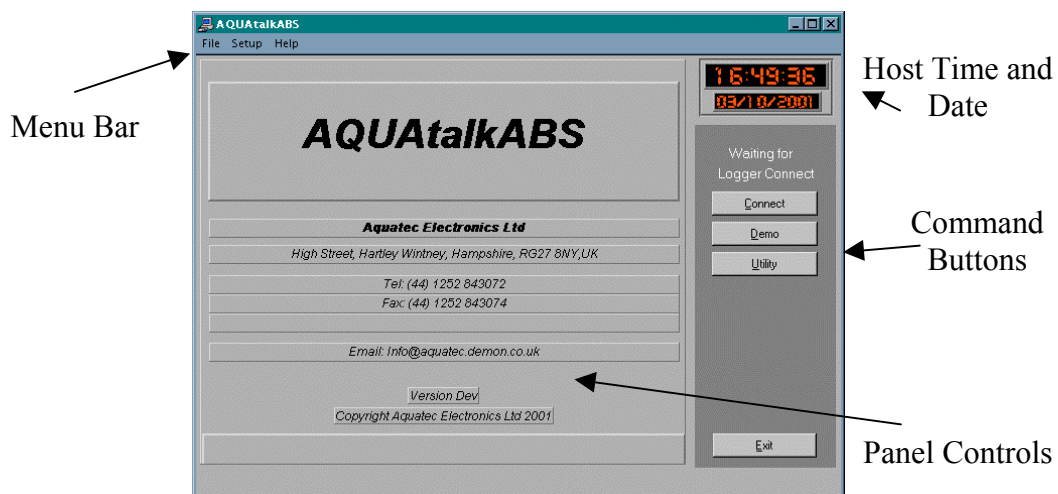


Figure 5 - Initial Screen

In the manual, selection by menus is shown in the form **Logger – New Regime**, etc.

The basic application screen is shown in the screen shot above. The important features are:

- ❑ The menu bar at the top of the screen allows navigation through the application to select the various configuration and control screens

- ❑ The host date and time displays the current PC values. It is important that this is correct as the logger (unless set to a specific user defined time) will use this time during all logging operations.
- ❑ The command buttons on the right of the screen give rapid access to the main screens (which can also be accessed through the menus).
- ❑ The actual panel controls for each of the different screens are used to display information or to request information from the user.

5.2.2 Initiating Communication

After running the AQUAtalk application and connecting the instrument to the computer via either the RS422 or USB the user is required to Connect to the AQUAscat ABS system, using either the **Setup – Connect** menu option or Connect button on the user interface. The AQUAtalk software automatically searches through all of the available communication ports on the host PC for a connected logger. On finding a system, the AQUAtalk application user interface will update, as shown in Figure 6.

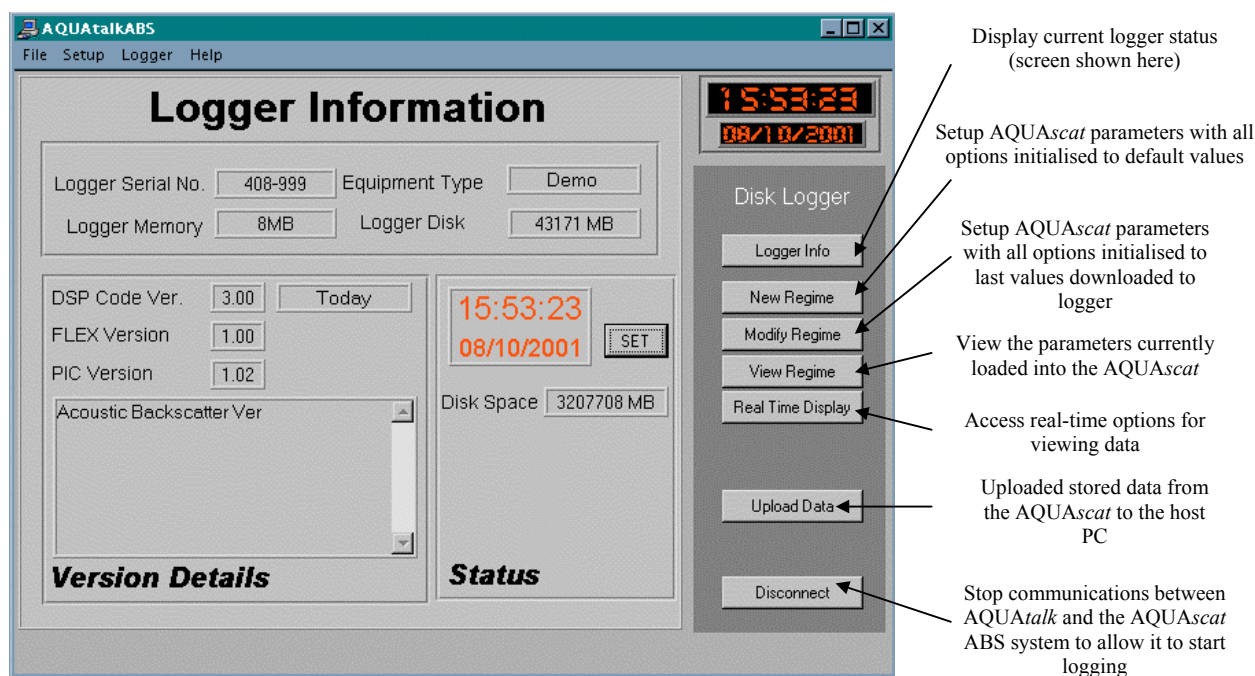


Figure 6 AQUAtalk display after connection

5.2.3 Modes

The AQUAtalk application can be operated in two modes: normal and advanced. Normal mode will allow the user to perform all of the operations required to setup and run the logger. Advanced mode provides access to low-level diagnostic functions. Unless under instruction from an Aquatec engineer, **DO NOT OPERATE IN ADVANCED MODE** as it may render the system inoperable.

5.2.4 Menu Options

Certain operations from the menu do not have an associated display panel as they do not require any data to be entered or displayed. These are:

- ❑ **File – Exit** to leave the AQUAtalk application
- ❑ **Setup – Connect** to connect the AQUAtalk application to the logger. If the logger is powered down, connecting to it (with **Setup – Connect**) will power it up on the RS232 connection.

Simply plugging the USB cable into the PC will also power up the system. If the logger was powered down, connecting will disable logging. If the logger was logging when the connection was made, the user will be prompted to end the current logging session (which will result in a truncated file) or leave it to complete and try again later. Once connected, the AQUAtalk application will interrogate the logger and display its current status.

- ❑ **Setup – Demo** – if the logger is unavailable, running in demo mode allows the user to operate most of the functions of the software as if a logger was attached.
- ❑ **Setup – Disconnect** to disconnect the AQUAtalk application from the logger. If the AQUAscet ABS system is connected via the RS232 port, about one minute after pressing Disconnect on the user interface, the logger will power down and be ready for logging. If the USB interface is used, the user should press Disconnect on the user interface and then remove the USB cable from one end of the link – the logger will then power down in about a minute.
- ❑ **Setup – Advanced** – allows the user to access the low-level diagnostic and test functions of the AQUAscet ABS System. This mode should normally **NOT BE USED**.
- ❑ **Setup – Settings** – allows the user to setup default directories for data uploaded from the system and firmware updates from Aquatec.
- ❑ **Help – About** – displays details of how to contact Aquatec. All questions should be directed here.

5.2.5 Menu Panels

The software operates through a series of panels, which form a subset of the display window. Each of these has one or more controls which may be used for data display or entry. The various controls have been selected to provide an intuitive user interface.

1. **Logger – Information** – displays information about the logger including how much disk space has been used and the current logger time.
2. **Logger – Upload Data** – allows the user to view and select which files to upload from the AQUAscet ABS system hard disk
3. **Logger – New Regime, Logger – Modify Regime** – both of these initiate the logger regime set-up wizard, which will take the user through all of the necessary screens to configure the logging regime. New Regime uses the application defaults as a starting point. Modify Regime uses the current logger settings as a starting point.
4. **Logger – View Regime** – displays the currently configured regime in the AQUAscet ABS system for the user.
5. **Logger – Format Disk** – allows the removal of ALL DATA from the disk. This process CANNOT BE UNDONE so ensure that all required data is uploaded before formatting.
6. **Logger – Settings** – configures which transducer is connected to which channel in the system. As any frequency or type of transducer can be attached to any channel, the user should check that these values are correct prior to deployment.
7. **Logger – AddIn Manager** – depending on the options purchased with the AQUAscet system, certain features may be disabled. Licenses can be purchased from Aquatec to enable certain features, as required. Contact Aquatec for details.

5.2.6 Menu Panels (Advanced Mode)

1. **Diagnostics – Flash Programming** – this allows the user to update the AQUAscet software with the latest version when supplied by Aquatec.
2. **Diagnostics – Crash Dump, Core Dump, Last Exception, Test Page, Disk Doctor, Tuning** – these should only be used to provide information for Aquatec engineers.

5.3 Connecting and Disconnecting a Logger

When AQUAtalk is initially loaded almost all functions are disabled until the user connects to the AQUAscat via either **Logger – Connect** or clicking on the Connect button. Connecting will automatically search for a logger connected on a USB port first and then on COM ports 1 to 4. Following a successful connection the client determines the state of the logger – if it is logging the user can terminate the current session, or wait for it to complete and connect later. The Logger Information screen is then loaded. It should be noted that while the PC is connected to the logger LOGGING WILL BE DISABLED unless the user explicitly initiates a logging session using the real-time controls. On a deployment, logging will not be re-enabled until the user has disconnected from the logger and it has powered down.

5.3.1 Powering up the Logger

The action of connecting a logger to an active RS422 or USB line causes it to power up, if it is not currently powered up for some other purpose. It will remain powered up for as long as the RS422 or USB connection is made and the line is active. The AQUAtalk software makes the line active after a successful Connect operation, and leaves it inactive after a Disconnect operation. The logger relies on time out periods, after which it suspends operation and powers down.

Note that for a USB connection the logger will remain powered up while the cable remains connected to the logger even if the PC is disconnected.

5.3.2 Connecting Another Logger

It is recommended that switching connections between two loggers not be carried out in mid-program, as this may lead to confusion. Instead, it is best to use the Disconnect button, connect the new logger, and press the Connect button again, ensuring that the most up to date information is held by the application.

5.3.3 USB Connection

If the USB cable is removed for any reason while connected then the driver will reset itself. Therefore you will need to reconnect to the logger (using the Disconnect then Connect). A loss of communications during a USB session may be due to a driver reset and so try reconnecting before anything else.

5.4 Examining and Changing Logger Parameters

Once communications have been established with the logger, AQUAtalk reads the logger's current status. The first panel displayed is the Logger Information panel.

5.4.1 Logger Information panel

The logger information panel shows information about the logger hardware. The screen is shown in Figure 7.

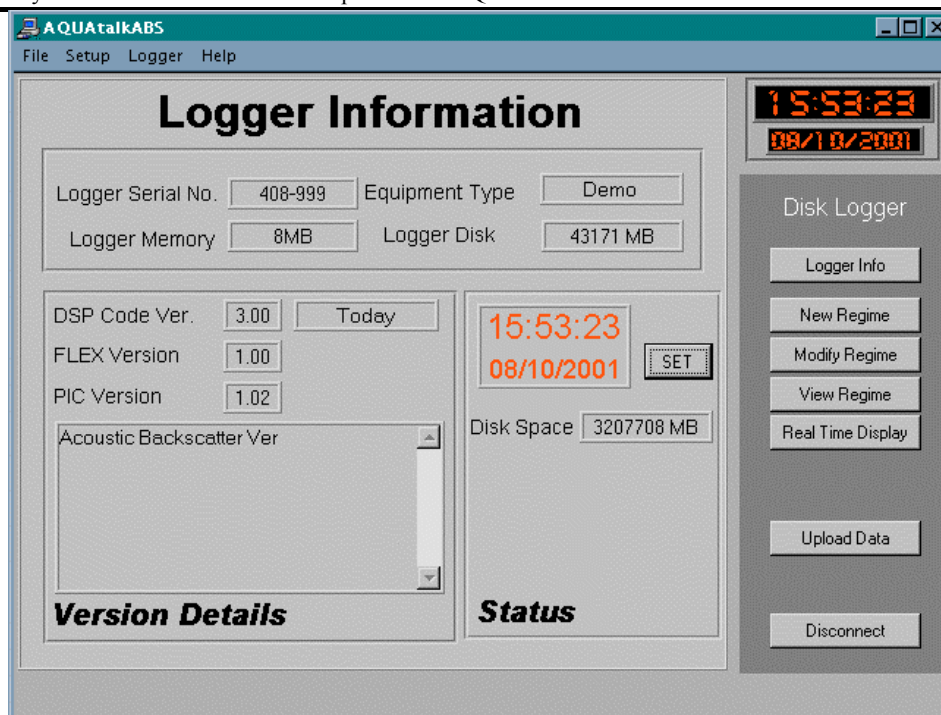


Figure 7 Logger Information Screen

The logger time displays the current time (updated once per second) held by the logger's internal real-time clock. This is the time used to time stamp all logging events so the user should ensure this is correct prior to deployment. The Disk Space control shows how much space (in Megabytes) has been used on the disk. The Logger Disk control shows how big the fitted disk is. The Version Details information shows the current versions of internal logger firmware and hardware – which should be supplied when requesting technical information from Aquatec.

5.5 Logger Parameters (Modify Regime)

The logger parameters govern the way in which the logger operates. Depending on the options purchased with your AQUAscat specific settings may be found in the Appendix. This subsection describes each parameter, including the possible values it may hold, and the default values. All of the parameters are setup via a wizard which takes the user through all of the steps required to setup the AQUAscat ABS system ready for a deployment. Modify Regime initialises all values to those currently set-up in the logger.

5.5.1 Transducer Selection

The transducer selection dialog determines the transducers attached to the system and sets the frequencies using information in the transducer.ini file, which is supplied with your system. You may also define how you wish to operate the transducer, this determines the dialog boxes that occur in the later sections of the wizard. All channels with the same type field will be configured the same way.

- **STANDARD** Is for ABS channels
- **SIDESCAN** Is to allow configuring as a side scan transducer
- **UPWARD LOOKING** Allows these channels to be configured differently to the Standard channels and therefore can be used for upward looking or different resolution.

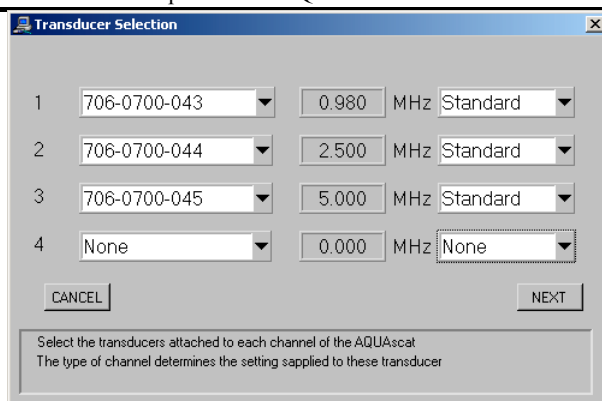


Figure 8 Transducer Selection

5.5.2 Logger Description

The first screen in the wizard is for the user to enter a description of up to 32 characters. This is provided solely for the user, and could be used for example, to store a location string of where the logger is to be deployed or an experiment name.

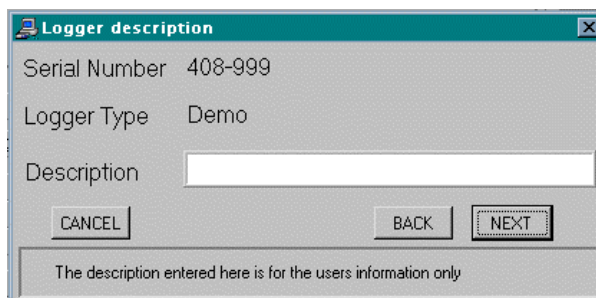


Figure 9 Logger description entry

5.5.3 Logger Time and Date

All of the files stored by the AQUAscat ABS system are time and date stamped. The time used is held in a real-time clock within the system – the time held in this clock is shown in red in Figure 10. If this is not correct the user can either synchronise this time with the host PC or enter an independent time by clicking on the Correct Logger Time button.

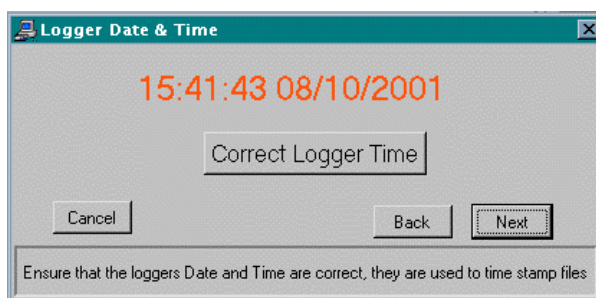


Figure 10 Logger Time and Date entry

5.5.4 Logging Regime

The logging regime screen defines how long and how often the AQUAscat ABS system should power up and log for. There are two modes of operation: Auto and Trigger.

5.5.4.1 Auto Logging

The Auto options are shown in Figure 11. In this mode the system will periodically wake up on a timer and log. If Instant Start is selected the logging will start on the first periodic timer event after the system has powered down. If Start At has been selected logging will start at the specified time and then continue on the periodic timer. When in this mode, the logger is also responsive to the external trigger line – it will wake from sleep on every trigger and log, in addition to the periodic wakeups.

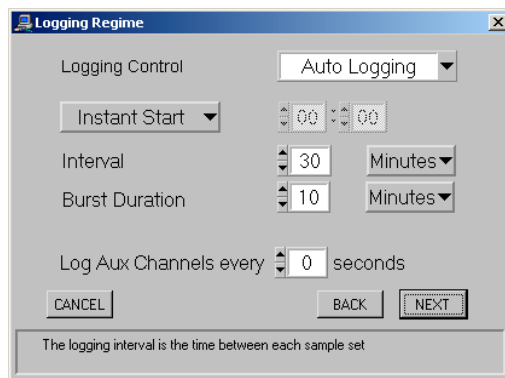


Figure 11 Logging Regime Auto Screen

The Interval allows the user to select how often the system should wake from sleep and log. It can be any value from 2 to 255 minutes. The Burst Duration sets the period the unit will log for once it has powered up and can be between 1 and 60 minutes. The user should note that the Interval must always be longer than the Burst Duration (or the logger would never power down) – the AQUAtalk application enforces range checking on these parameters. The File Extension option allows files stored on the logger to have a different file extension based on the data it contains. The default is (*.aqa).

- ❑ DEFAULT: 30 minute interval with 10 minute bursts from an instant start

5.5.4.2 Trigger Only

In trigger mode the wakeup of the logger is controlled completely by the EXTERNAL wake (EXT Wake) line supplied by an external system. The Log Every option selects after how many triggers the logger wake and log – it can be set to any value between 1 and 15.

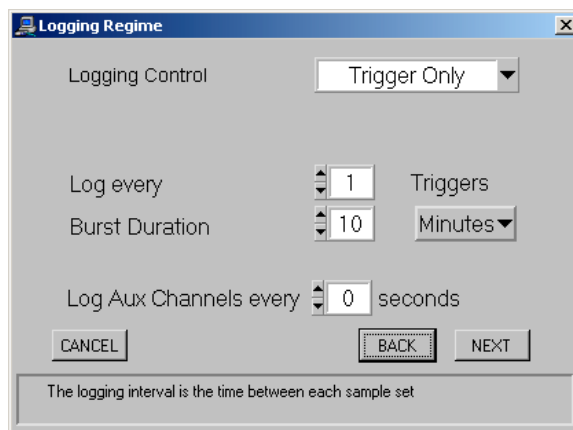


Figure 12 Logging Regime Trigger Screen

- ❑ DEFAULT: 10 minute logging every trigger

5.5.4.3 Auxiliary Channels

The sample rate of the auxiliary channels is controlled in the logging regime dialog by defining the sample interval for the channels in seconds. Thus the maximum rate is 1Hz. To disable the auxiliary channels set the rate to zero.

5.5.5 ABS Basic Settings

The basic settings define how often during a burst a profile is taken and how many profiles should be averaged together before storing the data. The profile rate defines how often the sequence of the enabled transducers should be fired. After each profile the data received can be averaged with other profiles to reduce the amount of data to be stored on the disk. The storage rate is calculated from the profile rate and averaging (storage rate = profile rate / averaging).

Depending on the licence purchased, the Complex Data and Profile Framing options will be enabled. If the complex data is selected, the data stored on the disk will have both magnitude and phase information. If this is selected, averaging is not possible so the data storage rate will be equal to the profile rate. The Profile Framing option allows the EXTERNAL trigger line (EXT Trig) to synchronise the AQUAscat with an external event. When imported into the AQUAview application, these triggers allow spatial averaging of triggered data sets.

- ❑ DEFAULT: 80Hz profile rate; 32 profiles per average

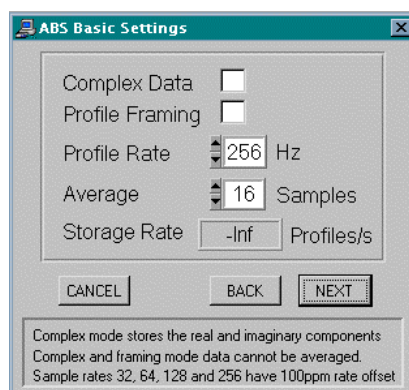


Figure 13 ABS Basic Settings screen

5.5.6 ABS Enable Channels

The standard ABS system is supplied with three transducers of different frequencies. However in certain circumstance (e.g. testing) the user may not want to have all transducers enabled to save storage space. This screen allows the user to select which transducers are actually enabled during a profile. It also displays which frequency transducer corresponds to which channel – it is important in deployment that the correct frequency transducer is physically connected to the correct connector on the logger housing. The resolutions available depend upon the licence purchased with the AQUAscat system. The standard system is supplied with a 10mm resolution. The High Resolution license allows additional resolutions of 5mm and 2.5mm for more detailed examination of smaller ranges. The Long Range license allows lower resolutions (20mm, 40mm, 80mm and 160mm) over extended distances for observations further into the water column. The resolution and distance range of channels on this screen effect all transducers defined as ‘Standard’ in the Transducer Setup screen.

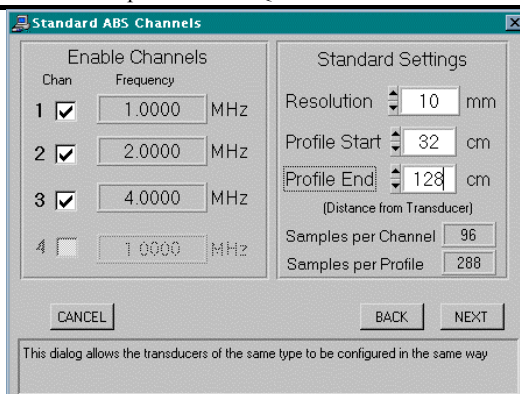


Figure 14 Enable Channels screen

5.5.7 Upward Looking Channels

Any transducers defined as 'Upward Looking' in the Transducer Setup screen can have an independent resolution and range to the 'Standard' transducers. The options are identical to those described in Section 5.5.6.

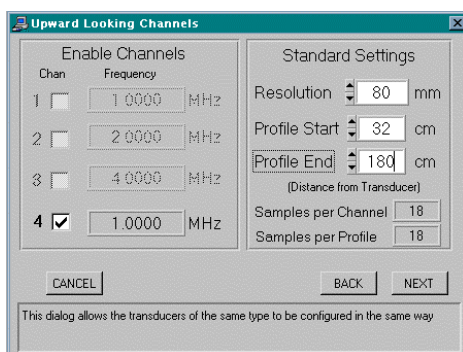


Figure 15 Upward Looking Channel Setup

5.5.8 Side Scan Setup

If any transducers are of the 'Side Scan' type (i.e. with a narrow wide beam), the Side Scan Setup screen allows control over the range and resolution settings. This type of transducer is particularly suited to observing ripple motion in the seabed, where it is placed at an angle to the bed.

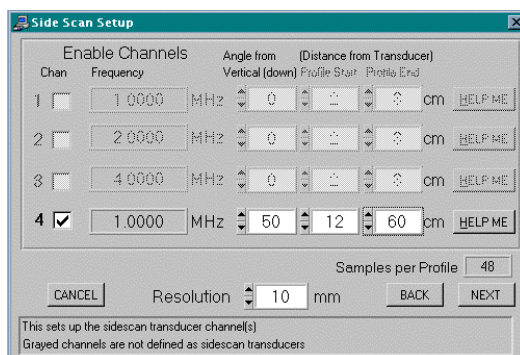


Figure 16 Side Scan Setup Screen

In this case the start and end distances correspond to a points along the seabed. The values are related to the angle of the transducer and its height above the bed. Pressing the HELP ME button

brings up a helper screen to calculate the required distances from the height and angle information. This screen is shown in Figure 17.

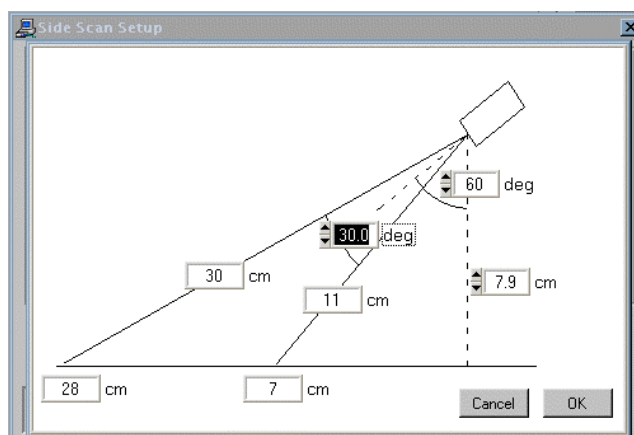


Figure 17 Side Scan Setup Help Screen

5.5.9 ABS Gain Settings

The receiver gain can be configured in several different modes, depending on the observation environment requirements. See section 3.2 for details on the gain control.

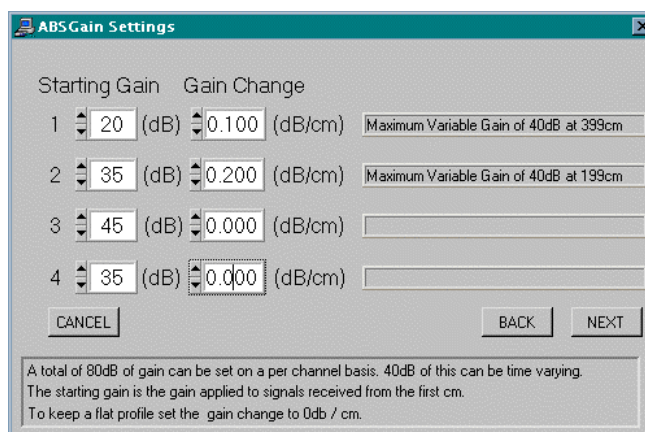
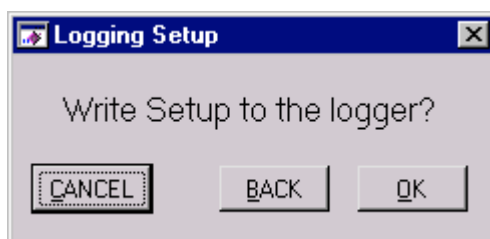


Figure 18 Gain Settings Screen

5.5.10 Logging Set-up

At the end of the wizard the user is prompted to write the set-up to the logger. 'OK' will download the set-up to the logger, overwriting the current values. 'Cancel' will abort, leaving the logger set-up unchanged and 'Back' allows the user to go back through the wizard and review the changes.

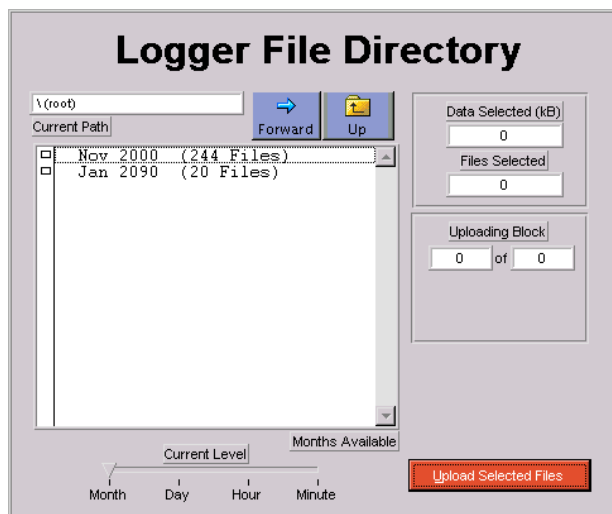


Note that logging will be held off until the logger is disconnected.

5.6 Recovering Data From the Logger

5.6.1 Logger Directory

Data may be recovered from the logger using the **Logger - Upload** menu option or clicking on the Upload Data button on the command bar. The upload panel is shown below. The user can navigate through the file structure by double clicking on an item in the list (or selecting the item and using the forward button). To move back up through the structure click on the Up button. The number of files contained within the category is displayed.



5.6.2 Selecting Files to Upload

To upload files to the PC the user must select individual files or groups of files by clicking on the square attached to the range to be uploaded. If all the files within a single month are required then select the month. This will automatically select all the files below. To select individual days, hours or minutes move down through the tree and select the required items. If only some items are selected lower down in the tree then an X is displayed in the higher levels. The total number of files selected and the approximate quantity of data are displayed.

5.6.3 Initiating an Upload

Once all the files have been selected the upload is initiated using the 'Upload Selected Files' button. The upload is made up of blocks of files; the number of these blocks depends on the date ranges requested and is not an indication of quantity.

The user is prompted for a directory to copy the files to. Ensure that the target disk has sufficient capacity for the data to be copied. The user does not need to enter a filename as all files have a filename made up of the date and time of the logging burst.

5.6.4 Upload Baud Rate

RS422 data uploads can be carried out at one of three baud rates 115200, 57600, or 38400. The logger will be automatically configured to use the rate selected. The higher baud rates may cause problems on slower computers due to the small amount of buffering within the PC. If an upload is aborting files or having more than 5 packet retries per file then use a slower baud rate.

The USB contains its own flow control scheme and therefore operates at maximum speed without user intervention.

5.6.5 Upload Progress

Progress of each file and the total progress are monitored on the progress bar. The user is provided with an approximate data size and a record of the number of files uploaded to date. The upload can be abandoned at anytime using the Abort upload button.

USB uploads are very fast so the file progress bar will appear to jump around randomly for small files. The currently uploading filename is also provided.

The user is informed of any error or warning messages that occur during an upload. The upload continues in the background unless the error aborts the progress.

An error log file is created in the upload directory and should be checked to ensure that all files were uploaded without error.

5.7 Deleting Data on the Logger

When all data has been recovered from the logger session, it is possible to clear the disk. This is done by reformatting the disk, and consequently is *irreversible*!

MAKE ABSOLUTELY SURE THAT ALL DATA OF INTEREST HAS BEEN UPLOADED.

Click on the **Format Drive** button on the button bar to format the disk. A progress bar is displayed at the bottom of the application window. This may take between 15 minutes and an hour, depending upon the disk size.

The drive should be reformatted prior to deployment to ensure that sufficient space is available for writing files.

5.8 AQUAtalk Data and Directories

When data is uploaded from the logger, it is stored in the directory specified by the user. File names comprise a long date in the form:

YYYYMMDDhhmmss.AQ*

where the file type extension is one of the following:

File extension	File type
AQA	AQUAscat file format

5.9 Real Time Data Display

Normally when the AQUAtalk application is connected to the AQUAscat, logging is disabled. However, using the real time display functions the system can be forced to initiate a logging session and display the data acquired in real time. There are two modes of real time operation: ABS Scrolling Data and Oscilloscope.

In both real time modes, the setup of the logger is as defined by the current regime held in the logger. The AQUAscat unit will operate as it would in a deployment situation – the user can view the acquired data in real time to rapidly assess the validity of the current regime (e.g. is the gain setting correct / is enough averaging being used). The regime can be quickly updated and the effect of the new settings examined. Clicking on Start Acquisition button will initiate the logging session

in both real-time modes. The user will be prompted with a Start Logging dialog box – the options are:

- ❑ Local Disk – the data is stored on the logger hard disk as well as being viewable on the PC in the real-time data windows. Following the experiment the file can be uploaded if required. The maximum file size for a file on the logger is 2Gbytes
- ❑ Log To PC – the data acquired is written to a file in the selected directory on the PC as well as being viewable in the real-time windows. Data is **not** written to the logger hard disk. This file will be named with the time at which the current session started. The maximum file size that can be stored in this mode is 2Gbytes.
- ❑ No Logging – the acquired data is only viewable in the real-time windows and is **not** stored anywhere.

The acquisition will continue for the period defined by the burst duration, unless the Stop Acquisition button is pressed.

5.9.1 Real Time Scrolling Display

The simplest display is the scrolling display. As each profile is acquired and processed by the AQUAscat, periodically the AQUAtalk application will request a profile to upload from the system. It is immediately uploaded to the PC and displayed on the intensity plot. The grey-scale plot is generated directly from the uploaded data points – white corresponds to zero sediment concentration, while black corresponds to full-scale sediment concentration. On the left hand scale of each display is the distance from the face of the transducer head.

Only the channels enabled in the regime are displayed – channels not enabled will not be displayed. The 'Window Size' button allows the user to run the display either in the AQUAtalk window (default) or full screen for a larger viewing area.

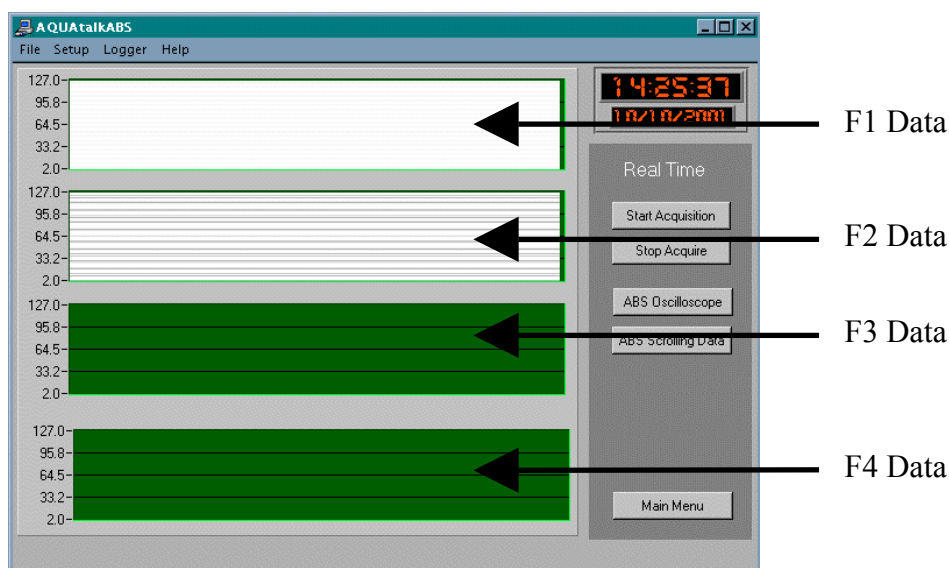


Figure 19 Real Time Scrolling Display screen

5.9.2 Real Time Oscilloscope

On this display the data acquired by the AQUAscat is plotted as it would be on an oscilloscope. Along the x -axis is the distance from the transducer head (in cm) and the y -axis displays the absolute sample value.

The display will not update unless the 'Update On' checkbox is ticked. When this is selected the display is periodically updated with the latest profile. The scaling of the data on the display is selectable to either 'Magnitude' where the data displayed as the actual 16 bit point value that is acquired by the system [0..65535] or 'Log' where the value displayed is $20\log_{10}(\text{sample value} / 2^{16})$ dB.

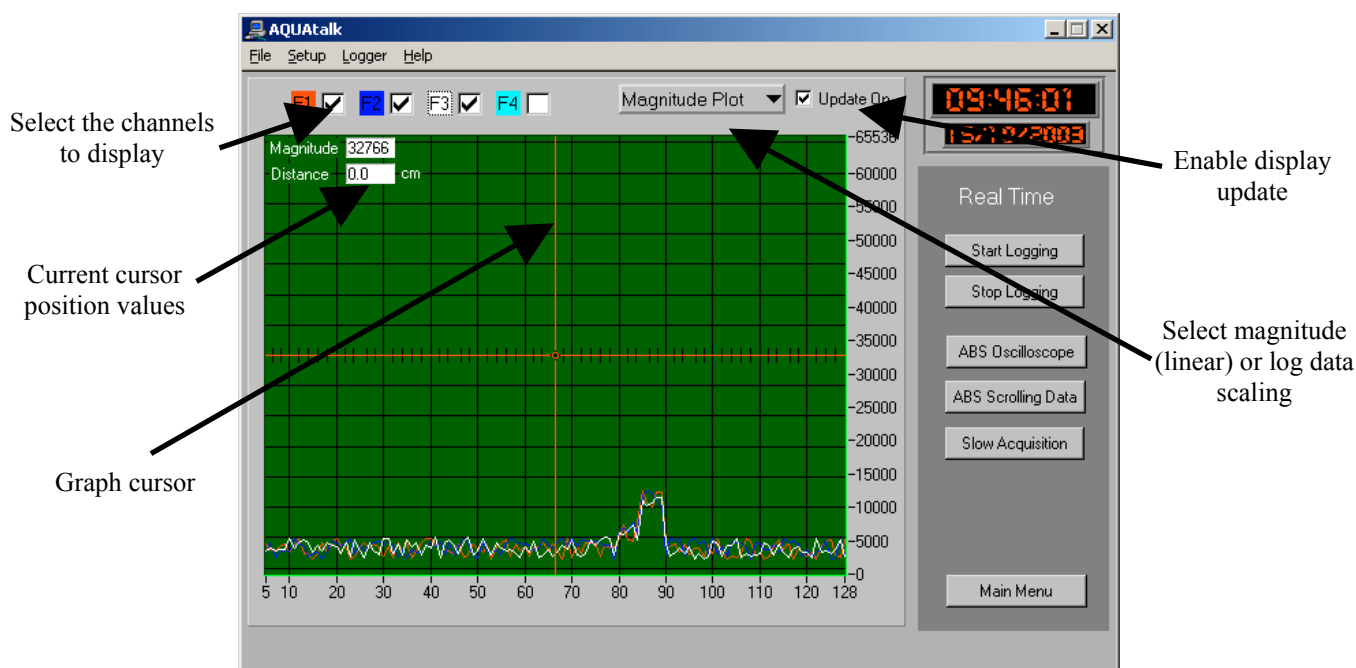


Figure 20 Real Time Oscilloscope display

On the display the red cross-hair cursor can be moved around the screen by holding the left mouse button down and dragging the cursor to the point of interest. The value of the point on the cross-hair is displayed in the two numeric boxes on the display.

6 Operation of AQUAview Software

6.1 Introduction

This section covers the general operation of the AQUAview software.

6.2 AQUAview Software Overview

AQUAview has been written to run under 32-bit Windows (98/2000/ME/NT) on an IBM PC or compatible. The software is designed to provide the user with a self-explanatory method of command entry, using simple keystrokes or mouse movements.

6.3 General Controls

The File menu at the top of the screen is used to select a default data directory, and to select files for analysis. When a file is selected, it is automatically read, and the first screen full of data is displayed. The display will appear as one, two, or three intensity plots, depending upon how many frequencies were logged.

A typical display is shown in Figure 21.

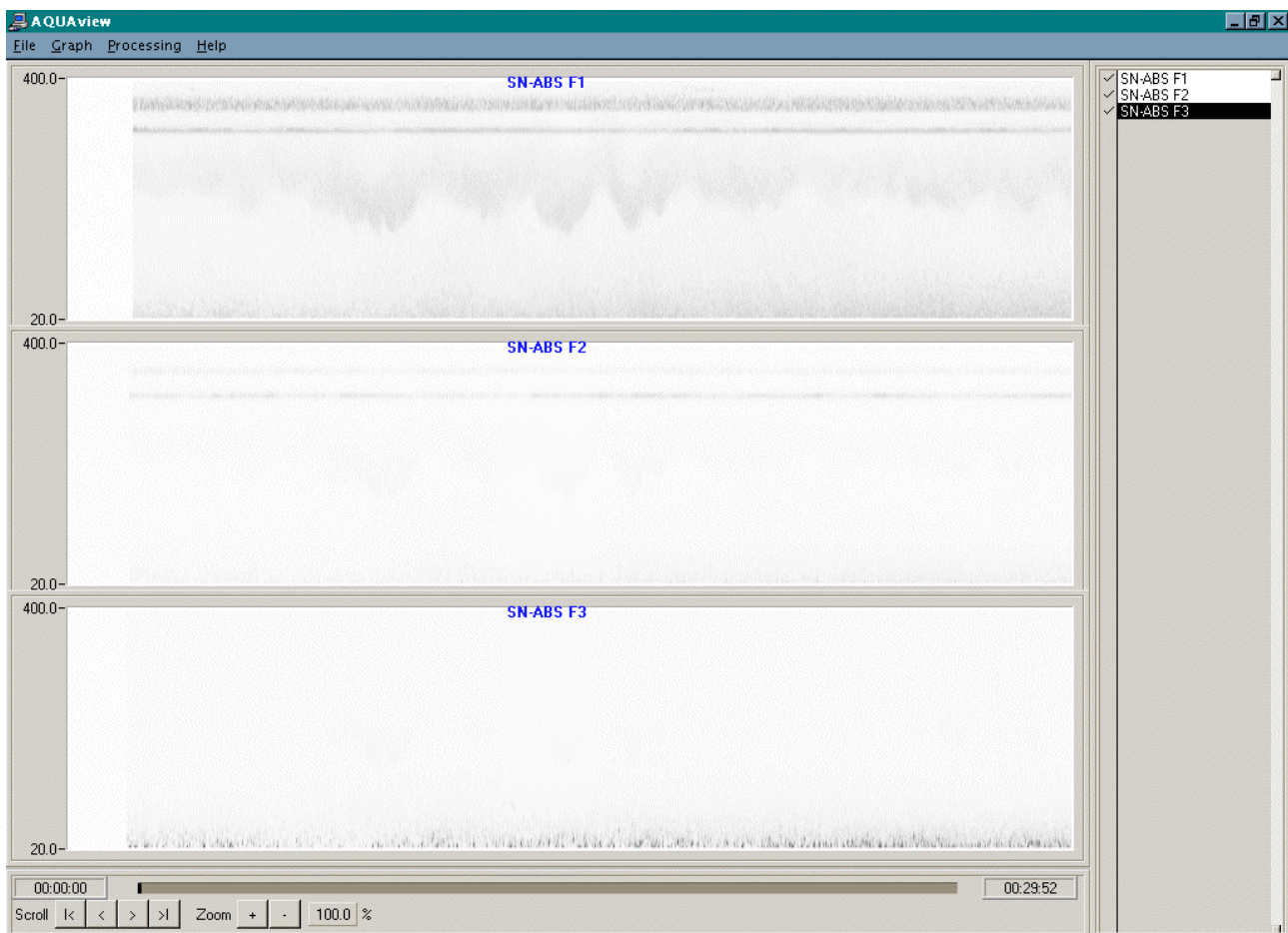


Figure 21 Typical AQUAview display

6.4 Program Controls

When the file is first loaded into AQUAview it is scaled so the entire file fills the screen. Initially no channels are selected, so the display is blank – the user must select which channels to display by clicking next to the channel name in the upper right corner of the screen. Once a tick appears by the side of the channel, its contents will appear on the screen. As more channels are selected, the displays scale so the whole screen area is always utilised. The time range is displayed at the lower left corner of the display (start time of the data) and at the lower right corner (end time of the data). By clicking on the ‘+’ and ‘-’ zoom controls a more detailed view of the data can be observed. Once the Zoom display is greater than 100% the scrolling control arrow buttons allow stepping through the data (forward ‘>’, back ‘<’, to start ‘|<’ and to end ‘>|’). Also the slide bar at the bottom of the screen can be dragged left and right to scroll through the data.

6.5 Menus

6.5.1 File Menu

Import – select the AQUAscat file to load

Export – export the currently loaded data to a CSV (comma separated variable file) for import into any spreadsheet application. When this is selected, a menu will popup to allow the selection of which channel to export.

Default Directory – sets the default directory to import files from

6.5.2 Graph Menu

Interpolate – adds additional points to the displayed data, to make it appear less blocky. This feature only alters the visual display of the data.

Remove All – clears all currently enabled channels from the display.

6.5.3 Processing Menu

These options perform functions on the selected data set, and generate a new channel containing the processed data.

7 Technical Specification

Frequencies	Up to 4 frequencies in the range 1 MHz to 6 MHz, typically 1 MHz, 2.5 MHz, and 5 MHz
Transducers	Typically Ø10mm ceramic discs (beam width according to frequency), with 12 mm and 5 mm versions available (for narrower or wider beam width).
Gain	4 stages of amplification are present in the system. Software controlled gain profiles can be downloaded for either linear TVG or fixed gain – varied as standard at 1 cm intervals
Range	128 cm standard. Other options available.
Transmitted Signal	1W, 13µs pulse standard. Other options available.
Transmission Rate	80 Hz max pulse rate for each frequency (i.e. 240 pulses per second for 3 frequencies).
Data Averaging	Groups of 1, 2, 4, 8, 16, 32 or 64 profiles are averaged to give a maximum output data rate of 80 Hz.
Range Selection	A window of configurable length is defined in terms of a start and end distance from the face of the transducer head.
Logging Period	From 1 to 60 minutes
Trigger	Either external hardware trigger when required or internal software trigger at regular intervals.
Automatic Logging	The logger can be set to take data at a fixed internal rate between 2 minutes and 255 minutes.
Fore/Background	A digital input or logger parameter determines the tagging of data. This is typically used to distinguish between foreground (high activity) and background (low activity) events. This line may also be used to control instrument gain.
Power Requirement	Logger consumption typically 2.1W when logging, 3.5W when writing to disk, with standby current of 200µA when not logging.
Additional Channels	Up to four additional analogue channels may optionally be monitored.
Data Acquisition	16-bit measurement resolution 1cm standard spatial resolution up to 1.25mm spatial resolution optional
Memory	1Mb x 16, representing approximately 2500 averaged three-frequency profiles at 1cm resolution.
Disk Storage	Any 2.5" notebook style IDE hard disk – typically 6GB minimum.
Data Communication	RS422 up to 115kbaud USB interfaces typically 3-4Mbaud.

8 AQUAscat File Format (Revision 2.4)

8.1 Introduction

The AQUAscat uses a propriety binary file format. Information is written to the file in packets, which identify the source and format of the contents. Packets are written to the file in time order, however the order of different packet types cannot be guaranteed. The import filter must therefore search for the specific packet that it requires.

Each packet has a standard 6 byte header which allows the remaining packet contents to be read or skipped. The format of the header is shown below:

Type	Description	Format
unsigned short	Packet Type. Most significant 4 bits always 1010 v indicates the version of the specific packet type cc is the specific packet type ID (1-256)	1010vvvv cccccccc
unsigned short	Size of packet body in bytes	ssssssss ssssssss
unsigned short	Header Checksum (Packet type XOR body size)	cccccccc cccccccc
.....	Body is packet specific	

Note 1:MSB is used to mean most significant byte and LSB is least significant byte

Note 2:The data in the file is stored in Low Bytes, High Bytes order. This is standard for IBM compatible PC.

Variable Types used in the file format:

- Byte is an unsigned 8 bit integer
- Unsigned short is an unsigned 16 bit integer. (2 bytes)
- Short is a signed 16 bit integer (2 bytes)
- Long is a signed 32 bit integer (4 bytes)
- Double is a 64 bit floating point number (8 bytes)
- BCD is binary coded decimal. Each group of 4 bits represents a number from 0-9

8.2 File Version

The first packet within an Aquatec formatted file should be a File Version packet. This describes the version of the file content and allows the import filter to reject unsupported files.

Packet Type	PAC_FILE_TYPE	Type constant (decimal)	19
Type	Description	Format	
unsigned short	File type (FILE constants)	number (see below)	
unsigned short	Major Version	number (starting from 1)	
unsigned short	Minor Version	number (starting from 0)	

File Types

- FILE_ABS 0 File containing acoustic backscatter data

The file may contain data from other supporting instruments as defined by the AQUAscat information packet.

8.3 AQUAscat Information

The AQUAscat information packet contains details about the instrument's hardware and capability. *Note that this packet is only included in file version 3 and above.*

Packet Type	PAC_AQUASCAT_DETAILS	Type constant (decimal)	45
Type	Description	Format	
unsigned short	Instrument Type	number	
unsigned short	Serial Number	number	
unsigned short	Board Mod Level	number	
unsigned long	Memory Size (in MB)	number	
unsigned long	Disk Size (KB)	number	
unsigned long	Disk Space (KB)	number	
ASCII (14 char)	Instrument Type (ASCII) string only terminated if less than 14 characters	"ABS Logger"	
BCD (2 byte)	Build Date Month/Day	0xMMDD	
BCD (2 byte)	Build Date Year	0xYYYY	
unsigned short	DSP Code Version (Major (M) + Minor (m))	MMMMMMMM mmmmmmm	
unsigned short	FLEX Code Version (Major (M) + Minor (m))	MMMMMMMM mmmmmmm	
unsigned short	PIC Code Version (Major (M) + Minor (m))	MMMMMMMM mmmmmmm	
unsigned short	Number additional Boards/Instruments	number	
unsigned short	Board 1 Type Code	type code INSTRUMENT_	
unsigned short	Board 1 Version	MMMMMMMM mmmmmmm	
ASCII (12 char)	Board 1 Description string only terminated if less than 12 characters	"Turbidity"	
unsigned short	Board 2 Type	type code INSTRUMENT_	
unsigned short	Board 2 Version	MMMMMMMM mmmmmmm	
ASCII (12 char)	Board 2 Description...	"Turbidity"	
	etc for each additional board/instrument		

Board/Instrument Types:

- INSTRUMENT_ABS 1
- INSTRUMENT_GPS 2
- INSTRUMENT_TURBIDITY 3
- INSTRUMENT_TEMPERATURE 4
- INSTRUMENT_PRESSURE 5
- INSTRUMENT_SALINITY 6
- INSTRUMENT_ADC486 8

8.4 Logging Session Packet

The logging session packet indicates the start date and time of the acquisition.

Note that the instrument type and serial number fields are only valid in File Version 3 and above. The entries in these fields should be ignored in previous versions.

Packet Type	PAC_LOGGER_HEADER	Type constant (decimal)	20
Type	Description	Format	
unsigned short	Instrument Type	number	
unsigned short	Instrument Unit Number	number	
unsigned long	Burst Number since regime written	number	
BCD (2 bytes)	Start Time (S=seconds)	0xSS00	
BCD (2 bytes)	Start Time (H=hour M=minute)	0xHHMM	
BCD (2 bytes)	Start Date (M=month D=day)	0xMMDD eg 0x0604=4 June	
BCD (2 bytes)	Start Date (Y=year)	0xYYYY eg 0x2001	
unsigned short	Auxiliary Channels Burst Interval	Seconds (0 = off)	
unsigned short	RESERVED	not used	
unsigned short	Wake Source		

The complete instrument serial number can be made up by producing an ASCII string of the instrument type and unit number (type-unit e.g 408-33).

The wake source indicates what caused the logging session. The following are possible:

- PERIOD_WAKE 4 when internal timer woke system up
- COMMS_WAKE 3 when user triggered logging
- EXTERNAL_WAKE 6 when an external trigger woke system up

8.5 Exception Packet

Any error occurring within the logger is written to the file. The exact location of the error will not be known. This is intended to act as a warning mechanism. Aquatec can provide an error.c file for the displaying of error messages from the error code.

Packet Type	PAC_ERROR	Type constant (decimal)	29
Type	Description	Format	
short	Aquatec Error Code (see error.h)	negative number	
short	Parameter	error code specific	

8.6 ABS Regime Packet (version 2)

The updated packet is used in file version 3 and above. This has been modified to deal with longer ranges.

Packet Type	PAC_ABS_PARAMETERS	Type constant (decimal)	21
Type	Description	Format	
unsigned short	Binary flag (F) for each channel indicating if it is on	0000 FFFF	
unsigned short	Profile rate in Hz	number	
byte	Averaging	number	
byte	Averaging (described as Power of 2)	0=average 1, 1=average 2 2=average 4etc	
unsigned short	Control Flag bits	Bit 0 Hardware Framing Bit 1 Complex (CH0-3) Bit 2 Gain Mode (Auto/Manual) Bit 3 FUTURE USE Bit 4 Complex (CH5-8) Bit 5 Auto Track Bottom Bit 6-15 FUTURE USE	
unsigned short	FUTURE USE		
unsigned short	FUTURE USE		
unsigned short	FUTURE USE		
unsigned short	Number of channels in following section	number	
unsigned short	Tx Pulse Length (steps of 104ns)		
unsigned short	Decimation	0=1.25mm, 1=2.5mm, 2=5mm 3=10mm 4=20mm 5=40mm etc	
unsigned short	Start Distance (cm)		
unsigned short	Stop Distance (cm)		
unsigned short	Time between this frequency TX and next frequency TX	(approx. ms)	
short	transducer angle (if entered correctly by user)	degrees	
byte	(scaled gain for logger)		
byte	Gain Step every x cm	number of cm per step	
short	Fixed Gain (in dB)		
float	Gain step in (dB *100)		
unsigned short	RESERVED		
unsigned short	RESERVED		
	Frequency 2 settings as Frequency 1		
	Frequency 3 settings as Frequency 1		
	etc for number of frequencies indicated		

8.7 ABS Magnitude Data Packet

The data in this packet has had its magnitude calculated from the real and imaginary data collected within the logger. Each packet contains a single profile for each of the enabled channels. If averaging is enabled the data in this packet will be the result of that process.

The content of the data packet must be determined from the PAC_ABS_PARAMETERS. The number of samples in each profile can be calculated from the start, stop and decimation information for each channel.

$$\text{ChannelSamples} = (\text{stop} - \text{start}) / 2^{(\text{decimation}-3)}$$

The channel samples must be calculated for each enabled channel. If a channel is disabled there is no data available for it.

Packet Type	PAC_ABS_DATA16	Type constant (decimal)	22
Type	Description	Format	
unsigned short*N	N Data values for first enabled channel	0-65535	
unsigned short*N	N Data values for next enabled channel	0-65535	
	etc		

8.8 ABS Complex Data Packet

The complex data packet contains non averaged real and imaginary components for each point in a channel's profile. The packet contains a single profile for each channel. As with the magnitude data the content must be determined from the PAC_ABS_PARAMETERS.

Packet Type	PAC_ABS_DATA_COMPLEX	Type constant (decimal)	41
Type	Description	Format	
(2 * short) * N	N real and imaginary data pairs for first enabled channel	0 to ±32767 (real) 0 to ±32767 (imaginary)	
(2 * short) * N	N real and imaginary data pairs for next enabled channel	0 to ±32767 (real) 0 to ±32767 (imaginary)	
	etc		

8.9 ABS Information packets

The ABS packets are for general information regarding events that occur with the ABS system. These include the logging of framing triggers and exception messages. The first value in the body of the packet indicates the type of information, the rest of the content then depends on this.

ABS information packet types:

- ABS_PROFILE_TRIGGER 9

8.9.1 Profile Trigger Packet

Packet Type	PAC_ABS_INFORMATION	Type constant (decimal)	44
Type	Description	Format	
unsigned short	ABS_PROFILE_TRIGGER	9	
unsigned short	seconds since start of logging	number 0-65535 seconds	
unsigned short	milliseconds since start of logging	number 0-999 ms	

Note: Timecode is not present in this packet until file version 3.

8.10 ABS Transducer Information

The transducer information provides details about the transducer frequencies and beam patterns.

Packet Type	PAC_ABS_TRANSDUCER	Type constant (decimal)	40
Type	Description	Format	
unsigned short	Number of channels		
unsigned short	Transducer Type	TRANSDUCER_type	
unsigned short	Shape constant	AQSHAPE_type	
unsigned short	Transducer diameter (vert. size if sidescan)	mm	
unsigned short	Transducer horizontal size (mm)	mm	
float	Transmit Frequency	Hz	
float	Receive Frequency	Hz	
char[16]	Transducer Serial Number		
	Frequency 2 as frequency 1		
	Frequency 3 as frequency 1		
	Frequency 4 as frequency 1		

Transducer types are described below:

- TRANSDUCER_ABS 1 Transducers defined for backscatter use
- TRANSDUCER_SIDESCAN 2 Transducers defined for sidescan use
- TRANSDUCER_SINGLE 3 A transducer defined for other use
- AQSHAPE_CIRCLE 1 Cylindrical Transducer
- AQSHAPE_RECTANGLE 2 Rectangular Transducer

8.11 Instrument Data

This packet is intended to store data from a variety of low rate instruments. The content is shown below:

Packet Type	PAC_INSTRUMENT	Type constant (decimal)	46
Type	Description	Format	
unsigned short	Instrument type	INSTRUMENT_XXXXX	
unsigned short	Quantity Including (timecode)	Quantity	

unsigned short	Timecode (second)	Seconds
unsigned short	Timecode (millisecond)	Milliseconds
	array of data values format depends on type	

- INSTRUMENT_WAVEC Wavec Bouy Data Packet
- INSTRUMENT_ADC486 8 Channel ADC Data Packet
- INSTRUMENT_TURBIDITY Turbidity channels

The INSTRUMENT_ADC486 packet contents are defined below:

Packet Type	PAC INSTRUMENT	Type constant (decimal)	46
Type	Description	Format	
unsigned short	INSTRUMENT_ADC486	INSTRUMENT_xxxxx	
unsigned short	10	Quantity	
unsigned short	Timecode (second)	Seconds	
unsigned short	Timecode (millisecond)	Milliseconds	
unsigned short	Channel 1	Millivolts	
unsigned short	Channel 2	Millivolts	
unsigned short	Channel 3	Millivolts	
unsigned short	Channel 4	Millivolts	
unsigned short	Channel 5	Millivolts	
unsigned short	Channel 6	Millivolts	
unsigned short	Battery Voltage	Millivolts	
short	Temperature	1/10 degree	

8.11.1 Turbidity Instrument

Data in Turbidity instrument packet is as follows, the total length of the packet excluding the timestamp is 20 bytes.

Packet Type	PAC INSTRUMENT	Type constant (decimal)	3
Type	Description	Format	
unsigned short	INSTRUMENT_TURBIDITY	INSTRUMENT_xxxxx	
unsigned short	22	Quantity	
unsigned short	Timecode (second)	Seconds	
unsigned short	Timecode (millisecond)	Milliseconds	
unsigned short	Turbidity 1 Gain	number	
Float (IEEE)	Scaled Turbidity reading 1	number	
unsigned short	Turbidity 2 Gain	number	
Float (IEEE)	Scaled Turbidity reading 2	number	
unsigned short	Turbidity 3 Gain	number	
Float (IEEE)	Scaled Turbidity reading 3	number	
unsigned short	Turbidity 4 Gain	number	
Float (IEEE)	Scaled Turbidity reading 4	number	
Float (IEEE)	Scaled Temperature reading external	number	
Float (IEEE)	Scaled Temperature reading internal	number	
Float (IEEE)	Not used (channel 7 on ADC)	number	
Float (IEEE)	Scaled battery voltage	number	

9 Sample Tutorial

9.1 Introduction

This appendix gives a brief description of a sample tutorial session, which is provided to allow the user to run through typical operations with the AQUAscat ABS System and the associated AQUAtalk Software package.

9.2 Session Objectives

The objectives of the tutorial session are:

- ☐ To connect a logger to the AQUAtalk software and establish communications
- ☐ To set up the logger with the following parameters:
 - ☐ Auto-Logging;
 - ☐ Interval = 5 minutes;
 - ☐ Logging Period = 1 minute
 - ☐ Sample Rate = 80Hz, Averaging = 16 profiles per average (=>Storage Rate = 5Hz)
 - ☐ Frequencies = F1 and F3 only
 - ☐ Range = 5-128 cm
 - ☐ Fixed gain = 50dB; Gain change = 0dB per cm
- ☐ To set the logger time;
- ☐ To ensure space is available for logging;
- ☐ To start logging;
- ☐ After two sets of data has been recorded, to end logging;
- ☐ To examine what data has been recorded;
- ☐ To recover the logged data;
- ☐ To view the logged data;

9.3 Getting Started

First ensure that the hardware and software are installed correctly, as described in the earlier sections of this manual. Then run the AQUAtalk application.

Press the Connect button. The Logger Information panel should be displayed. Verify that the logger serial number indicated on the screen matches that of the connected logger. The program is now ready to proceed.

9.4 Checking and Making Space on the Logger

In this tutorial, only two data sets are to be recorded. The space taken by this may be calculated as follows:

Space = StorageRate x Number of Frequencies Enabled x Number of cm in the range x BurstDuration (in seconds) x Number of Bursts x 2 (bytes per value) x Overhead (approx. 1%)

In this case:

Space = 5 x 2 x (128 – 5) x 60 x 2 x 2 x 1.01 = 298152 bytes (approx.)

Select **Logger – Information** to examine the current logger data store. The Disk Space value indicates the number of MB of FREE SPACE on the disk. Check that there is at least 1MB free. If

not, then after ensuring that the data on the disk is not valuable, select the **Logger - Format Disk** menu item to erase all data from the disk. Note that this takes some time.

9.5 Setting up the Regime

Select **Logger – New Regime** – this will initiate the regime set-up wizard.

- ❑ On the Logger Description, enter any description – “Test setup” if you’re short of ideas! Click NEXT.
- ❑ On the Logger Data & Time check the time on the logger (in red text). Click Correct Logger Time, then on the Logger Time dialogue click on PC Time. The AQUAscat and host PC times are now synchronised. Click NEXT.
- ❑ In the Logging Regime dialog select Auto Logging, with the logging Interval set at 5 minutes and the Burst Duration set at 1 minute. Leave the File Extension set on Default(.aqa). Click NEXT.
- ❑ Now we will setup the ABS Basic Settings. Set the Profile Rate to 80Hz and the Average to 16 samples – this will give us a storage rate of 5 profiles per second. Leave the Profile Framing and Complex Data unchecked (if available). Click NEXT.
- ❑ On the ABS Enable Channels dialog select (with the tick box) Chan 1 and 3 (which on a standard AQUAscat will correspond to nominally 1MHz and 5MHz respectively). Set the Resolution to 10mm, the Profile Start to 5cm and the Profile End to 128cm. Click NEXT.
- ❑ On the ABS Gain Settings dialog select the Fixed Gain at 50dB on Chan 1 and Chan 3. Leave the Gain Change at 0dB per cm and the Mode in Standard Mode. Click NEXT.
- ❑ The Logging Setup dialog will now appear. Click OK to download the new regime to the logger.
- ❑ A Logger Start Time dialogue box will now appear – the time shown is the time the AQUAscat ABS system will start logging PROVIDED THE SYSTEM HAS POWERED DOWN BEFORE THIS TIME.

The logging regime that you have just entered should now be displayed. On the selected regime the first 5-minute interval is timed from the instant when the new regime is downloaded. Take a note of the current time on the AQUAtalk display your first data will be logged 5 minutes from now.

9.6 Logging

Click on the Disconnect button to end communications between the AQUAtalk application and the AQUAscat ABS system. If you are connected using the USB interface, you should now also disconnect the USB cable from one end of the link. Within one minute the logger will power down into its sleep state. If you have taken more than four minutes from downloading the regime to disconnecting the logger, the AQUAscat will not have time to go to sleep before waking up for the first 5 minute alarm – therefore the first alarm will be ignored and the first data will be logged 10 minutes from the regime being downloaded.

In a lab setup, the user can see if the logger has gone to sleep as its current consumption will drop below 600uA (from a 12V supply). Now leave the system alone for about quarter of an hour after the noted time. This will allow the second data set to be sampled set to be taken and written to the

disk and the system may be about to gather a third set. Reconnect to the system. If the logger is currently logging, a popup box will be displayed to inform of this fact – do not stop logging when prompted; wait for one minute and try to connect again.

9.7 Recovering Logged Data

Select **Logger – Upload**. A file list will appear allowing selection of the desired files. Select the desired files as described in Section 5.6.2 – the file that have just been logged should be the most recent 2 or 3 files. Once all files required have been selected, upload using the **Upload Selected Files** button. You will be prompted for a directory to store the uploaded files in.

9.8 Viewing Logged Data

Once the data has been successfully uploaded, run *AQUAview* to examine the data. Select **File – Select File** and pick one of the files that were uploaded. The first 128 profiles on each of the two frequencies will be displayed on the screen. You can also examine and confirm the Logger info and Logger setup, and the File information. For more information about viewing the files, see Section 6.

10 Fault Finding / Correction

10.1 Introduction

This section is intended to provide elementary fault diagnosis and rectification information. In the event that this does not provide a solution to the apparent problem, contact Aquatec with full details of the fault symptoms.

10.2 AQUAtalk program will not communicate with logger

The AQUAtalk program runs, but never establishes a connection with the logger.

10.2.1 General

Ensure the external power supply to the system is turned on. The supply should be between 10 and 28V dc. At 12V the supply current will typically be between 100 – 200mA when the disk is inactive and between 200mA and 300mA when the disk is busy. On a power up from sleep the hard disk spin-up may momentarily peak at over 500mA – the user should ensure the power supply used can supply sufficient power while remaining within the specified limits. If a variable voltage supply is used, raising the voltage will reduce the current demand.

10.2.2 RS422 Serial Cable Problems

Check that the serial cables are undamaged, and that they are securely connected at both ends and at any intermediate connection points. Ensure that the supplied RS422 converter is securely attached between the host PC RS232 port and the AQUAscat ABS system. Only use the supplied RS422 converter, as other converters will most likely have a different pin-out.

10.2.3 USB Cable Problems

Check that the USB cables are undamaged, and that they are securely connected at both ends and at any intermediate connection points. Under Windows, when the AQUAscat ABS system is plugged into the host PC it should be found by the hardware manager – if the AQUAscat has never been used on the system before, you should be prompted for a driver for the hardware – see Section 4.

10.3 AQUAscat did not log when expected

After the new regime has been downloaded the AQUAtalk application MUST be disconnected from the logger, so it can power down before it will first start logging on its first wake up. If the USB interface was used to connect to the logger, it MUST BE PHYSICALLY UNPLUGGED to allow the logger to power down.

10.4 AQUAtalk program will not upload data

If all other communications are operational but data uploads are unreliable and causing files to be cut short then reduce the upload rate from 115200 to 57600 or 38400.

Packet retries are not a problem. However if they occur more than five times in a file it may be more efficient to use a slower baud rate.

10.5 Files do not contain all expected data

Logging will terminate before the time specified by the Burst Duration in the regime if:

- ❑ the user has logged the file in the real-time mode and pressed the Stop Logging button before the burst duration has expired.
- ❑ the user connects to the AQUAscat while it is currently logging, the AQUAtalk application will inform the user of this condition and allow logging to be disabled – this will result in a truncated last file.

10.6 Logger time is wrong after I have set it and disconnected

The logger is normally fitted with a clock battery backup, so if the external power supply is removed its time should be maintained. However, if the battery goes flat, and the logger is disconnected from its power supply for any reason then the real time clock time is lost. Replace the battery.

11 Appendix – Optional External Channels

11.1 Turbidity and Temperature

If the AQUAscat has been ordered with external turbidity and temperature channels the AQUAtalk application will look slightly different to the screen shots shown in the previous sections.

11.1.1 Temperature Channels

There are two channels of temperature – one is external and mounts on the end of a cable plugging into the 2-pin connector on the AQUAscat housing and the other is mounted onto one of the electronics PCB boards within the housing. The external probe is specified to $\pm 0.5^{\circ}\text{C}$ in the range 5 to 25 $^{\circ}\text{C}$. The internal sensor accuracy is not specified (typically within $\pm 2^{\circ}\text{C}$). The temperature channels can be read via the Real Time display screen. When logging, if the turbidity channels are enabled then the temperature channels will be logged – otherwise they are not logged.

11.1.2 Battery Voltage Channel

The supply voltage to the AQUAscat is monitored and can be logged – it will only be logged if the turbidity channels are enabled.

11.1.3 Turbidity Channels

The turbidity channels in the system use Seapoint Turbidity sensors – a separate manual is included specifically relating to these sensors. Each of the 4 sensors in the system can be individually configure to one of 4 gain modes, or placed in auto gain. The individual gain ranges are described in the Seapoint manual. In auto gain mode the logger takes a sample at each of the 4 settings and determines which gain setting is best for the current conditions. Therefore in less cloudy conditions a higher gain will be selected allowing a higher resolution measurement to be made.

The linearity of the Seapoint sensors in the different gain modes is good, so a measurement taken at one gain will read approximately the same number of FTU as in another gain. There are some points to note:

- ❑ When changing the gain the Seapoint sensor requires 1 second for its internal filters to settle, limiting the maximum sample rate to 1Hz in fixed gain.
- ❑ In auto gain the gain must be set to each of the 4 gains with a second between each to allow the gain to settle, so the maximum sampling rate to allow for some overhead is 6 seconds.
- ❑ When fixed gain is used, if the turbidity in the observation environment exceeds the range for the particular gain setting the sensor output will saturate and the value recorded will not represent the actual turbidity at that point in time – the actual output for the Seapoint sensor is 0.5V, but typically the output gets up to a value slightly below this, so for example when on the 25FTU range the actual maximum output of the saturated sensor could be 24.5FTU. This will be obvious in the data as a ceiling if a graph is plotted.

11.1.4 Real Time Display

In addition to the standard buttons on the Real Time command button panel, a Slow Acquisition button is available to view the current state of the turbidity and temperature channels.

The screenshot shows the AQUAtalk Data Acquisition window. It features a menu bar with 'File', 'Setup', 'Logger', and 'Help'. The main display area is titled 'Data Acquisition' and shows real-time data for four turbidity sensors, two temperature sensors, and supply voltage. Each sensor reading is in a red box. To the right of the turbidity readings is a 'Gain' column. On the far right, there is a 'Real Time' section with a digital clock and date, and several control buttons: 'Start Logging', 'Stop Logging', 'ABS Oscilloscope', 'ABS Scrolling Data', 'Slow Acquisition', and 'Main Menu'. At the bottom, a 'Time Code' is displayed as '00:00:43 20040427135130.aqa', and a 'Stop Logging' button is present.

Sensor	Reading	Unit	Gain
Turbidity 1	22.2	FTU's	5
Turbidity 2	25.2	FTU's	1
Turbidity 3	20.1	FTU's	1
Turbidity 4	20.0	FTU's	1
Temperature Internal	26.62	C	
Temperature External	20.98	C	
Supply Voltage	11.86	V	

Depending on the turbidity settings in the regime, the display will update once every second if fixed gain has been selected for all turbidity channels or once every 6 seconds if any channel has auto gain selected.

11.1.5 Additional Regime Settings

The screenshot shows the 'Turbidity Setup' dialog box. It has a title bar with a close button. The 'Turbidity Enable' checkbox is checked. The 'Sample Interval' is set to 6 seconds. Below this, there are four rows for 'Turbidity Sensor 1' through 'Turbidity Sensor 4', each with a dropdown menu for gain settings. The gain settings are: 5 * Gain, 1 * Gain, Auto Gain, and 1 * Gain respectively. At the bottom, there are 'CANCEL', 'BACK', and 'NEXT' buttons. A note at the very bottom states: 'Gain Settings for each Turbidity Sensor' and 'The minimum sampling interval for Auto gain is 6 seconds'.

In the regime settings an additional panel will be presented to setup the turbidity channels. The user can independently setup the gain mode for each of the channels and setup the sampling interval. If any channel is on auto gain, the maximum sample rate will be once every 6 seconds – otherwise it will be a maximum of every 1 second.

NOTE: For the temperature and battery voltage to be logged the turbidity channels MUST be enabled (even if they are not fitted).